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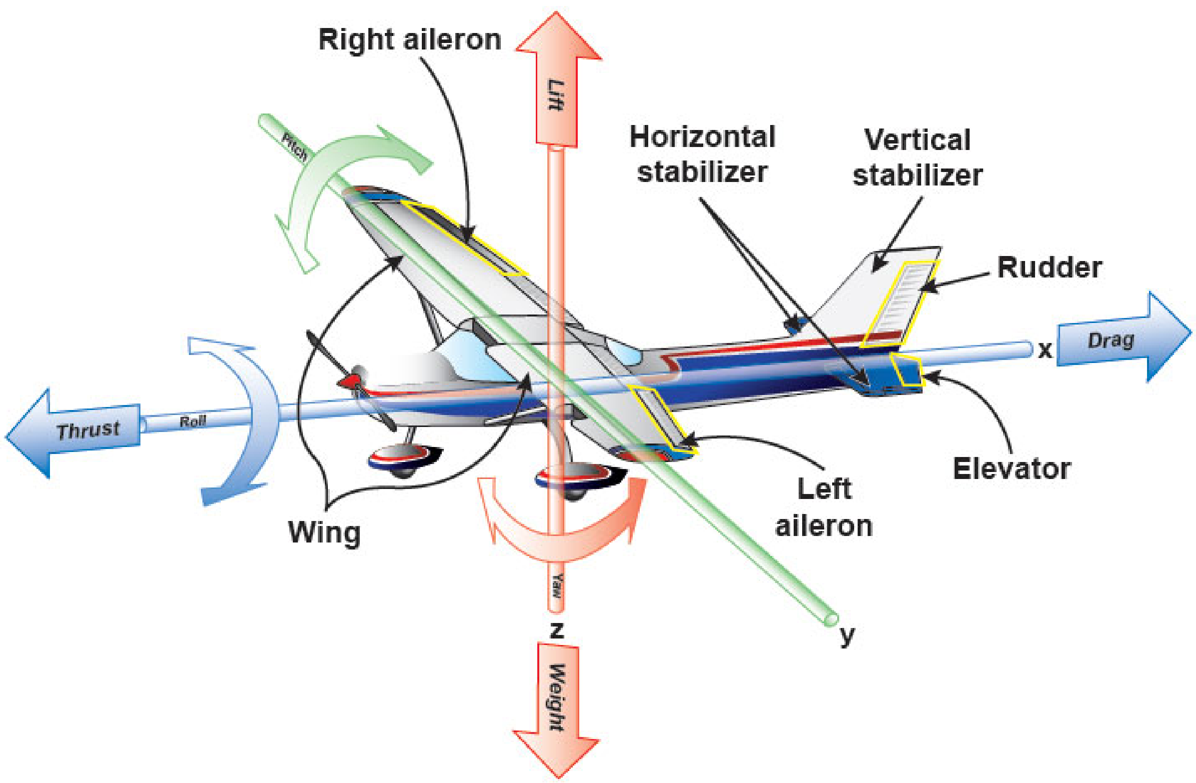

Part 66 Module 8 for A1/B1/B2 (Common)

BASIC AERODYNAMICS

Issued Oct 2012/Reviewed. September 2023

AMENDMENT RECORD

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Reviewed	Feb 2020	Alan Broadfoot MISTC	Full reformat with all new pictures
1	September 2023	Philippos Georgiou	Adaptation to combine A1/B1.1/B2 course
2	August 2024	Philippos Georgiou	Hish speed airflow as chapter 8.4
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MODULE 08 – BASIC AERODYNAMICS

Part 66 Ref

08.01 Physics of the Atmosphere

International Standard Atmosphere (ISA), Application to Aerodynamics.

08.02 Aerodynamics

Airflow around a body; Boundary Layer, Laminar and Turbulent Flow, Free Stream Flow, Relative Airflow, Up-Wash and Downwash, Vortices, Stagnation; The Terms: Camber, Chord, Mean Aerodynamic Chord, Profile (Parasite) Drag, Induced Drag, Centre of Pressure, Angle of Attack, Wash In and Wash Out, Fineness Ratio, Wing Shape and Aspect Ratio; Thrust, Weight, Aerodynamic Resultant; Generation of Lift and Drag: Angle of Attack, Lift Coefficient, Drag Coefficient, Polar Curve, Stall; Aerofoil Contamination including Ice, Snow, Frost.

08.03 Theory of Flight

Relationship between Lift, Weight, Thrust and Drag; Glide Ratio; Steady State Flights, Performance; Theory of the turn; Influence of Load Factor: Stall, Flight Envelope and Structural Limitations; Lift Augmentation.

08.04 High Speed AIrflow

Speed of sound, subsonic flight, transonic flight, supersonic flight, Mach number, critical Mach number, compressibility buffet, shock wave, aerodynamic heating, area rule; Factors that affect airflow in engine intakes of high-speed aircraft; Effects of sweepback on critical Mach number.

08.05 Flight Stability and Dynamics

Longitudinal, Lateral and Directional Stability (Active and Passive),.

Extra Reading

EASA Module 11A - Amazon Books

EASA Module 08 - Amazon Books

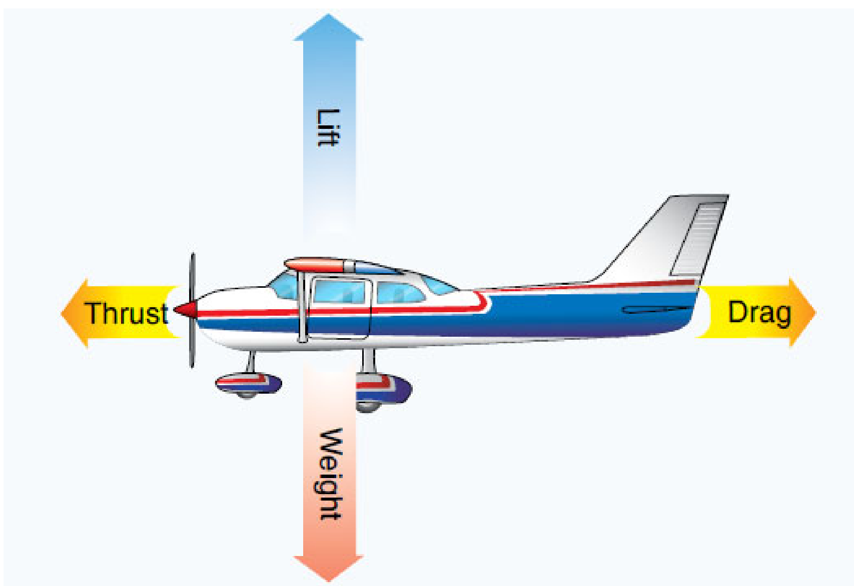
Jeppeson A & P Airframe textbook

AMT Mechanical Technicians Handbook

Dale Clark - Dictionary of Aircraft Terms

SECTION 08-01

PHYSICS OF THE ATMOSPHERE



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PHYSICS OF THE ATMOSPHERE

Introduction

The earth is encased in a thin layer of gases which insulate us from the devastating effect of the sun's energy and which supports life. This layer of gases, called our atmosphere, surrounds the earth to a depth of about 800km (500miles).

It consists of a mixture of gases, principally Nitrogen - about 78%, and Oxygen – about 21 %. The remainder 1 % is made up of traces of argon, neon, krypton and carbon dioxide.

The gases of our atmosphere are compressible and in the lower levels are pressed down upon by all of the air above. So, pressure and density are increased.

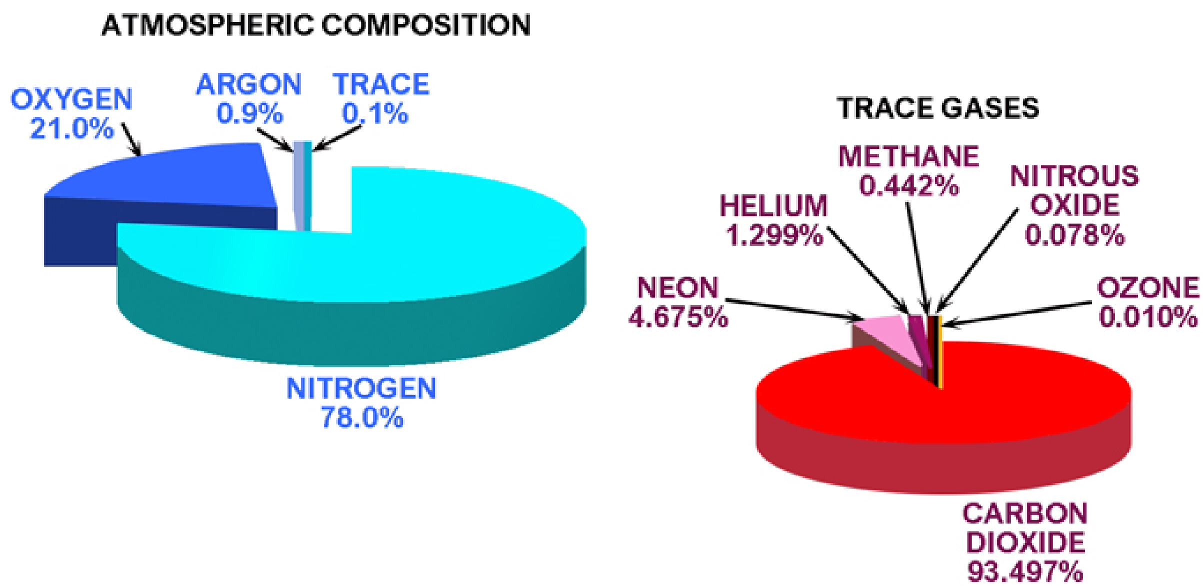

Fig 01. 1 - Atmospheric Composition

Troposphere

The lower level of our atmosphere is called the troposphere, and it extends from the surface upward to the tropopause. Its depth varies from about 7’500m (-28'000 feet) at the poles to 20'000m (-54'000ft) at the equator. The troposphere contains water vapour that causes clouds and vertical air currents and accounts for much of what we call weather.

Tropopause

This is the boundary between the troposphere and the stratosphere, and it is here that the temperature drop with altitude stops, and there is a constant temperature of -56.5°C.

Stratosphere

Above the troposphere the air is extremely thin and since there is no water vapor in the stratosphere, there isn’t any weather.

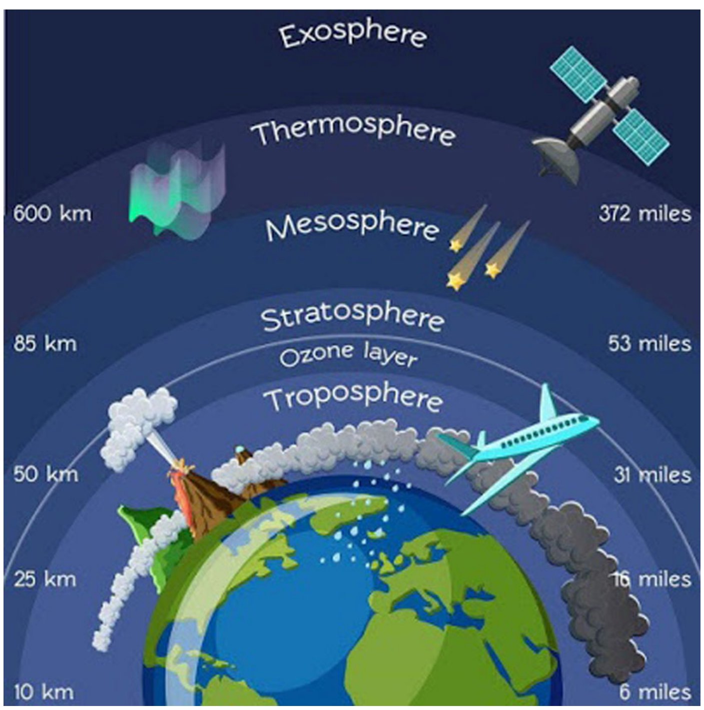

Fig 01. 2 - Atmosphere Layers

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ATMOSPHERIC CONDITIONS

Pressure

The weight of air above any surface produces a pressure at that surface i.e. a force of so many Newton's per square meter of surface. The average pressure at sea level due to the weight of the atmosphere is about 1013.25 millibars or 29.92 inches mercury.

The apparatus for measuring atmospheric pressure is called a Barometer. A glass tube open at one end and closed at the other is filled with mercury. The open end is sealed temporarily and then submerged into a small container partly filled with mercury, after which the end is unsealed. This allows the mercury in the tube to descend, leaving a vacuum at the top of the tube.

Some of the mercury flows into the container while a portion of it remains in the tube. The weight of the atmosphere pressing on the mercury in the open container exactly balances the weight of the mercury in the tube, which has no atmospheric pressure pushing down on it due to the vacuum in the top of the tube.

As the pressure of the surrounding air decreases or increases, the mercury column lowers or rises correspondingly. At sea level the height of the mercury in the tube measures standard day condition at 760mm (29.92 in).

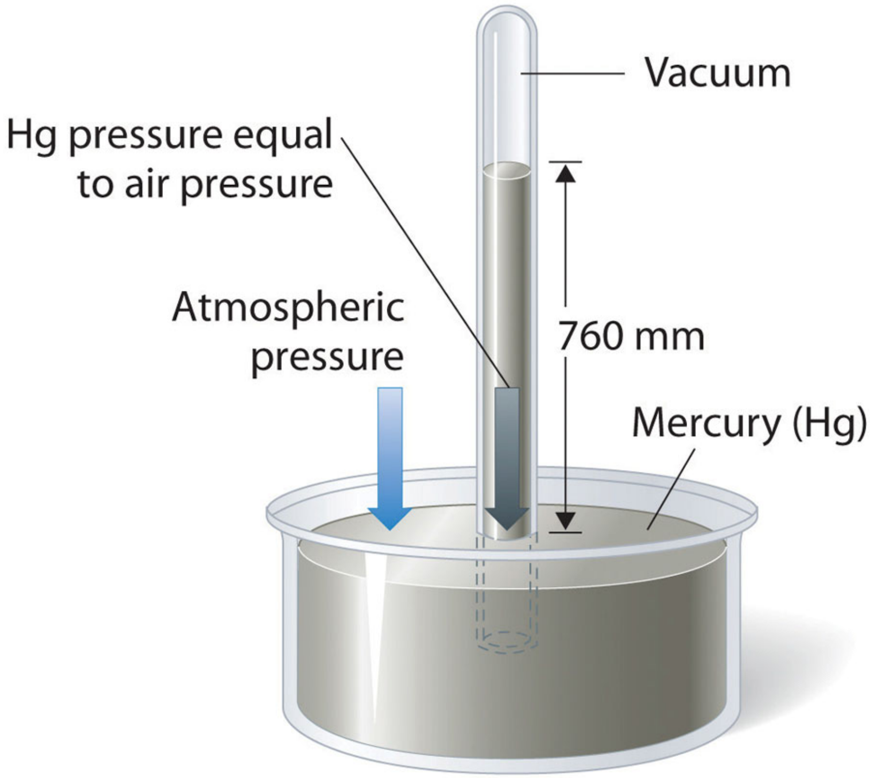

Fig 01. 3 - Mercury Barometer

Value	Unit	Explanation
1	bar	bar
1.033	Kg/cm²	Kilogram per sq centimetre
760	mm Hg	mm of Mercury
29.92	in Hg	inch of Mercury
14.7	psi	pound per sq. inch
1013.2	mb or hPa	millibar or hector Pascal

Fig 01. 4 - Table 1: Different Pressure Units

Density of Air

The density of air i.e. the mass per unit volume is low compared with water.

- **The mass of a cubic meter of air at ground level is roughly 1.225 kg**
- **The mass of cubic meter of water is a metric ton, 1000 kg, nearly 800 times as much).**

It is this very property of air, its density, which makes all flight possible, or perhaps we should say airborne flight possible.

The balloon, the airship, the kite, the parachute, and the aircraft, all of them are supported in the air by forces which are entirely dependent on its density; the less the density the more difficult flight becomes and for all of them flight becomes impossible in a vacuum.

The density of gases is governed by the following rules:

- **Density varies in direct proportion with the pressure (density increases as pressure increases)**
- **Density varies inversely with the temperature (density decreases as temperature increases)**

Air at high altitudes with low pressure is less dense than air at low altitudes with higher pressure. A mass of hot air is less dense than a mass of cool air.

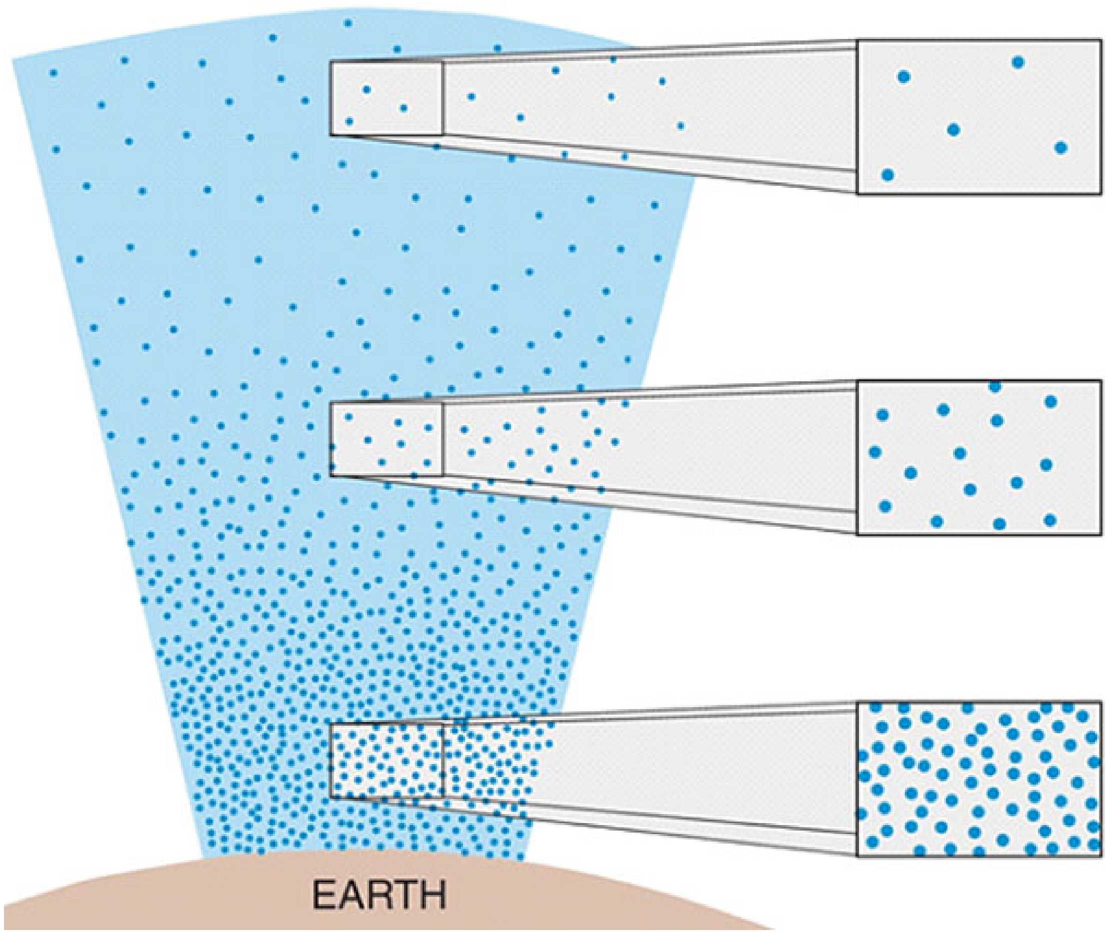

Fig 01. 5 - Air Density

Air Temperature

Going upward through the lower layers of the atmosphere there is the gradual drop in temperature. The reason for the falling off of temperature is that the radiant heat direct from the sun passes through the atmosphere without appreciably raising the temperature.

The earth, however, absorbs the heat, the temperature of the earth is raised and the air in contact with it absorbs some of this heat In still air, the temperature falls off with height at a rate of about 6.5°C for every 1'000m or 2°C for every 1 '000ft.

This rate of fall does not alter until about 11 '000m (36'000ft) - the actual height varying at different parts of the earth, being greater at the Equator (54'000ft) and least at the poles (28'000ft).

Air Temperature lapse rate= 6.5°C for every 1'000m (2°C per 1'000ft).

When the Tropopause is reached, quite suddenly the temperature ceases to fall and remains practically constant to the lower limits of the stratosphere.

Viscosity

An important property of air in so far as it affects flight is its viscosity. This means the tendency of one layer of air to move with the layer next to it; it is rather similar to the property of friction between solids.

NOTE:
For liquids, it corresponds to the informal concept of "thickness"; for example, honey has a much higher viscosity than water.

Fig 01. 6 - Viscosity (Low/High)

Humidity

Humidity is a condition of moisture or dampness. The maximum amount of water vapour that the air can hold depends entirely on the temperature; the higher the temperature of the air, the more water vapour it can absorb.

By itself, water vapour weighs approximately five-eight's as much as an equal volume of perfectly dry air.

Therefore, when air contains 5 parts of water vapour and 95 parts of perfectly dry air, it is not as heavy as air containing no moisture. This is because water is composed of hydrogen (an extremely light gas) and oxygen. Air is composed principally of nitrogen, which is almost as heavy as oxygen.

Assuming that the temperature and pressure remain the same, the density of air varies with the humidity. On damp days the density is less than it is on dry days.

For practical work in aviation, temperature and dew point are used more often than relative humidity to measure the amount of water vapour in the air. Dew point is the temperature to which a body of air must be lowered before the water vapour condenses out and becomes liquid water.

Temperature, Pressure, and Density versus Altitude

As altitude increases, both temperature and pressure decrease the former tending to increase air density and the latter to decrease it. One effect does not exactly compensate for the other because the pressure drop affects density more than the temperature effect.

Air Density and Aircraft Operations

Changes in air density affect the operation of aircraft regardless of whether the changed density results from changes in pressure, temperature, humidity, or any combination of these. If the air density drops there will be a reduction in:

- **The lift produced by wings.**
- **The thrust produced by propellers or jet engines.**
- **The power or torque produced by piston engines.**

The last two of these is significant when checking engine operation on the ground.

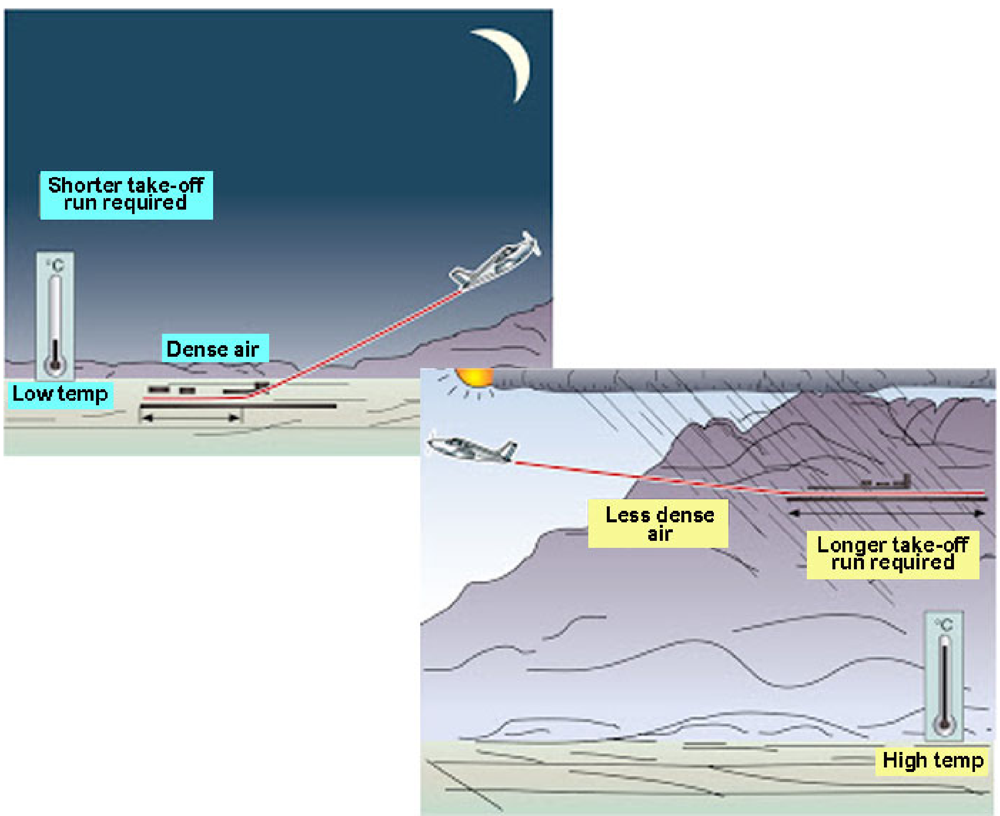

Fig 01. 7 - Temp Density Differences

THE STANDART ATMOSPHERE (ISA)

Since the temperature, pressure, and density of the atmosphere change from place to place and from day to day it is necessary to have a standard set of conditions with observed readings that can be compared to facilitate the correction of aircraft and engine performance calculations for any non-standard conditions.

The set of standard conditions is usually known as the International Standard Atmosphere (ISA). In the aircraft industry this is also known as the ICAO Standard Atmosphere. ICAO stands for International Civil Aviation Organisation.

The tables of Standard Atmosphere published by ICAO cover a wider range of altitudes, more accurately than the table shown.

If the performance of an aircraft is computed, either through flight tests or wind tunnel tests, some standard reference condition must be determined first in order to compare results with those of similar tests.

INTERNATIONAL STANDARD ATMOSPHERE				
ALTITUDE		DENSITY	PRESSURE	TEMP
FEET	METRES	(kg/m)	(inch Hg)	(°C)
0	0	1.225	29.92	15.00
1,000	305	1.112	28.9	8.50
2,000	610	1.007	27.8	2.00
3,000	914	0.9093	26.8	-4.49
4,000	1219	0.8194	25.8	-10.98
5,000	1524	0.7364	24.9	-17.47
6,000	1829	0.6601	24.0	-23.96
7,000	2134	0.5900	23.1	-30.45
8,000	2438	0.5258	22.2	-36.94
9,000	2743	0.4671	21.4	-43.42
10,000	3048	0.4135	20.6	-49.90
15,000	4572	0.1948	16.9	-56.50
20,000	6096	0.08891	13.8	-56.50

Fig 01. 8 - Table 2: The International Standard Atmosphere (I.S.A)

Variations from Standard Days

As may be expected, the temperature, pressure, density, and water vapour content of the air varies considerably in the troposphere. The temperature at 40 latitude may range from 50°C at low altitudes during the summer to -70°C at high altitudes during the winter.

As previous stated, temperature usually decreases with an increase in altitude. Exceptions to this rule occur when cooler air is trapped near the earth surface by a warmer layer. This is called temperature inversion, commonly associated with frontal movement of air masses.

Pressure likewise varies at any given point in the atmosphere. On a standard day, at sea level, pressure will be 29.92 in Hg. On non-standard days, pressure at sea level will vary considerably above or below this figure.

The density of the air is determined by the pressure and temperature acting upon it. Since the standard atmosphere can never be assumed to be "standard", a convenient method of calculating density has been devised.

Since air pressure is measured in inconvenient terms, it is expedient to utilise the aneroid altimeter as a pressure gauge and refer to the term "pressure altitude" instead of atmospheric pressure.

Pressure Altitude

Pressure altitude is the altitude in the standard atmosphere corresponding to a particular value of air pressure. The aircraft altimeter is essentially a sensitive barometer calibrated to indicate altitude in the standard atmosphere.

With the altimeter of an aircraft set at 1013hPa (29.92 in Hg), the dial will indicate the number of meters above or below a level where 1013hPa exist, not necessarily above or below sea level, unless standard day conditions exist.

In general, the altimeter will indicate the altitude at which the existing pressure would be considered standard pressure.

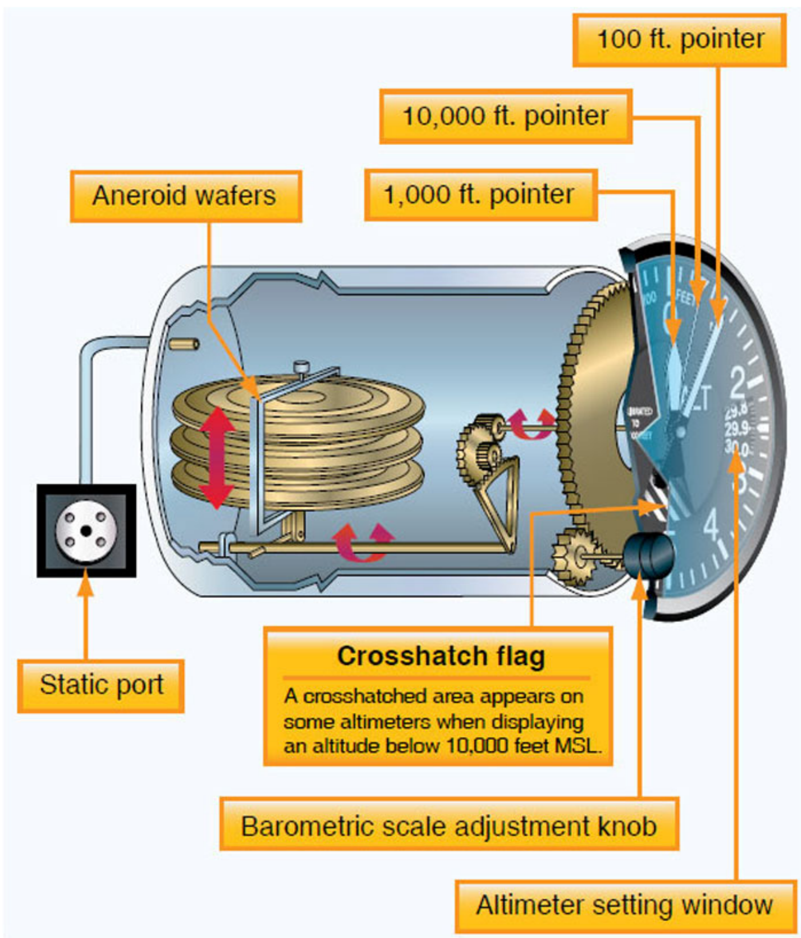

Fig 01. 9 - Aircraft Altimeter

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Pressure Altitude and Density Altitude

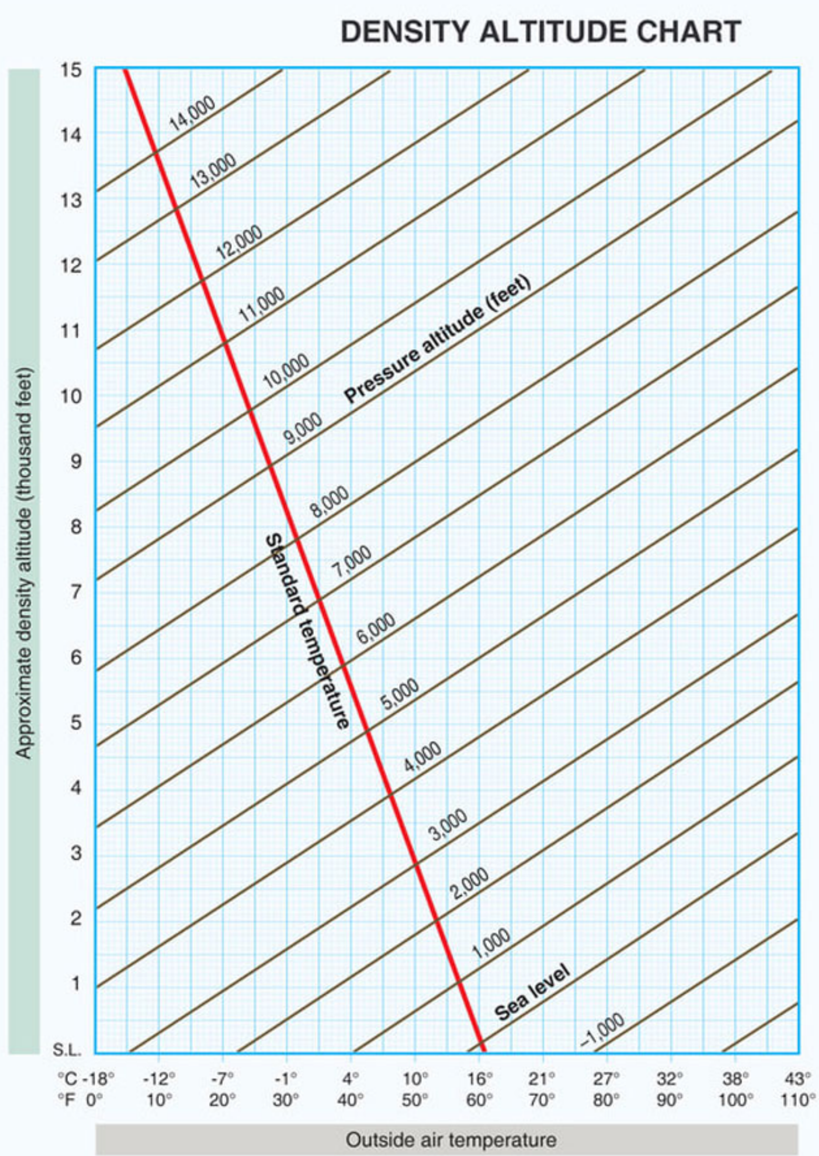
These terms relate to important aspects of aircraft operation and performance. Pressure altitude is the altitude listed for a given pressure in a table of the standard atmosphere.

Because of day-to-day and place-to-place variations of pressure from the standard, an aircraft's pressure altitude may differ from its true altitude. There is also the factor that the air density at any particular pressure altitude must be affected by temperature.

Density altitude is the altitude listed for a given density in a table of standard atmosphere. The density altitude will differ from the pressure altitude when the temperature is not standard.

There are no aircraft instruments that directly measure air density, but there are instruments that measure pressure altitude and air temperature. If these two factors are known, the air density can be found by reference to altitude conversion graphs.

These graphs could be used during flight, and they are also used in measuring the performance of engines during test runs on the ground.



Altimeter setting ("Hg)	Pressure altitude conversion factor
28.0	1,824
28.1	1,727
28.2	1,630
28.3	1,533
28.4	1,436
28.5	1,340
28.6	1,244
28.7	1,148
28.8	1,053
28.9	957
29.0	863
29.1	768
29.2	673
29.3	579
29.4	485
29.5	392
29.6	298
29.7	205
29.8	112
29.9	20
29.92	0
30.0	–73
30.1	–165
30.2	–257
30.3	–348
30.4	–440
30.5	–531
30.6	–622
30.7	–712
30.8	–803
30.9	–893
31.0	–983

Fig 01. 10 -Table 3 Density Altitude Chart

Equation of State

To calculate the values for the International Standard Atmosphere, up to the Tropopause, the Equation of State for a perfect gas is used. This equation is represented by the formula:-

$$p = pRT$$

where

p = Pressure millibars

p = Density kg/m^3

T = Temperature Kelvin

R = Gas constant (for air is 2.872)

Gas Laws

A perfect gas may be defined as a fluid which obeys Boyle's Law and the Laws of Charles. The so-called permanent gases, such as oxygen, nitrogen, hydrogen and also air under normal conditions, follow these laws fairly closely, and so for normal purposes, the various simple equations for a perfect gas may be applied.

Boyle's Law

States that if the temperature of a given quantity of a perfect gas is maintained constant, during a change in pressure and volume, the volume varies inversely as the pressure of the gas.

$$pv = constant$$

Charles' Law (Constant Pressure)

If the pressure of a gas is maintained constant the volume will increase by 1/273 of its value at 0°C for every 1°C rise in temperature.

$$\frac{V}{T} = constant$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Charles' Law (Constant Volume)

If the volume of a gas is maintained constant the pressure will increase by 1/273 of its value at 0°C for every 1°C rise in temperature.

$$\frac{P}{T} = constant$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Combined Gas Laws

The three gas laws, Boyle's, Charles' Volume and the Pressure Laws can be combined into one equation:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

By applying the above equation to atmospheric conditions using the ISA values then, providing the temperature and pressure at any altitude is known the density of the air can, within reasonable limits, be estimated for that stated altitude.

where:

$\frac{P_1V_1}{T_1}$ **are the sea level values**

And

$\frac{P_2V_2}{T_2}$ **are the altitude values**

Example

If the density of air at sea level is 1.225kg/m3 when the pressure is 1013mb and the temperature is 15°C (288K), find the density of the air at 8000m where the temperature is minus 37°C (236K) and the pressure is 357mb.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

so

$$V_2 = \frac{P_1V_1T_2}{T_1P_2}$$

And

$$\frac{1013*1*236}{288*357} \qquad New\ volume = 2.33m^2$$

Therefore 1m^3 of air at sea level expands to 2.33m^3 at 8000m but still has a mass of 1.225kgs. The density (mass/unit volume) at 8000m

$$\frac{1.225kg}{2.33m^2} = 0.526kg/m^3$$

By referring to Table 2 it can be seen that the assumed density under ISA values is stated as being **0.526 kg/m^3.**

AIRSPEED

Introduction

To study airflow physics, it is necessary to examine the measurement of airspeed, as the simple instruments in general use take no account of the difference between actual and ISA conditions.

As the conventional airspeed indicator is calibrated in accordance with the conditions of the ISA, it is a relatively simple matter to correct its reading to obtain a true value for airspeed in the conditions in which the aircraft is flying.

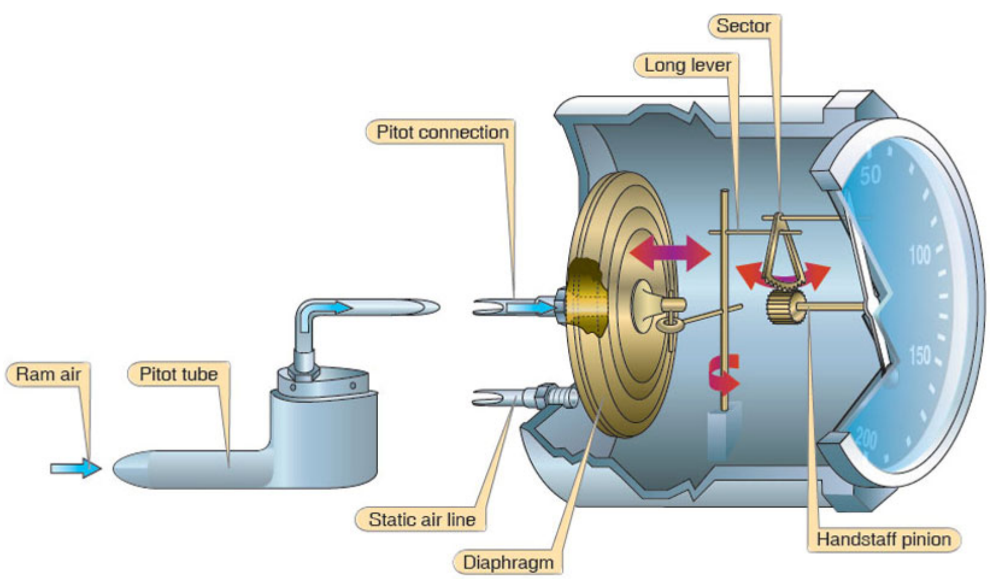


Fig 01. 11 - ASI

We have no need to study the airspeed indicator as part of this subject, but a simple explanation is given so that you may understand its principle.

ASI Principle

Anybody moving through the air will experience an increase in pressure due to the air piling up ahead of it. If we consider the body to be fixed and the air to be moving, then the pressure will be proportional to the Kinetic Energy of the air.

We should remember, from basic physics, that the Kinetic Energy of a solid object is obtained from the formula $\boldsymbol{E_K = 1/2MV^2}$. To calculate Kinetic Energy of air we must consider a unit mass, or the mass of a unit volume, that is 'Density'.

So the formula becomes $\frac{1}{2}\rho v^2$, where the Greek letter rho represents density.

This is the most important expression in aerodynamics, and will be examined further.

The ASI is supplied with the pressure due to the aircraft forward speed from the **PITOT** tube, and the pressure due to the aircraft height, **STATIC** pressure, and it compares the two. The result is shown on a pressure scale which has been converted to speed.

As the conventional airspeed indicator is calibrated in accordance with the conditions of the ISA, it is a relatively simple matter to correct its reading to obtain a true value for airspeed in the conditions in which the aircraft is flying.

AIRSPEED DEFINITIONS

Indicated Airspeed

Due to its simplicity, the Airspeed Indicator is not 100% accurate in its reading. Manufacturing tolerances and wear and tear in use result in errors of indication, which, if within limits, are considered to be acceptable.

The instrument is checked when new and at regular intervals. The errors for every point on the indicator scale are recorded and used as part of the calculation to obtain calibrated airspeed.

Calibrated Airspeed

The airflow patterns around an aircraft in flight will vary with differences in speed and altitude. The result will be that the pressure sensor (Pitot/Static tube), does not measure true pressure values.

As the variations will be the same for all aircraft of the same type, the manufacturer is able to obtain the true figures by experimental analysis. The resulting Pressure Error correction figures are published in the aircraft Flight Manual for use in making calibrated airspeed calculations.

The pressure error values, which apply to all aircraft of the same type, are mathematically combined with the instrument calibration values, which are specific to the instrument itself. The results are recorded on a correction card which is used by the pilot to make the conversion.

It should be noted that in British aviation, Calibrated Airspeed is commonly known as Rectified Airspeed, while in the USA it is known as True Indicated Airspeed.

Equivalent Airspeed

An accurate measure of dynamic pressure when the aircraft is flying fast. Air entering the pitot tube is compressed, which gives a false dynamic pressure **(IAS)** reading, but only becomes significant at higher speeds.

At a given air density, the amount of compression depends on the speed of the aircraft through the air. When the **IAS** is corrected for “position” and “compressibility” error, the resultant is **EAS**.

True Airspeed

To obtain an accurate airspeed the actual atmospheric temperature and pressure must be considered. As these factors are continuously variable, in flight, the pilot feeds the Calibrated Airspeed, Outside Air Temperature and Height, (which is pressure), into a calculator.

True Airspeed is then indicated, but it should be understood that it is only true at the time the calculation is made.

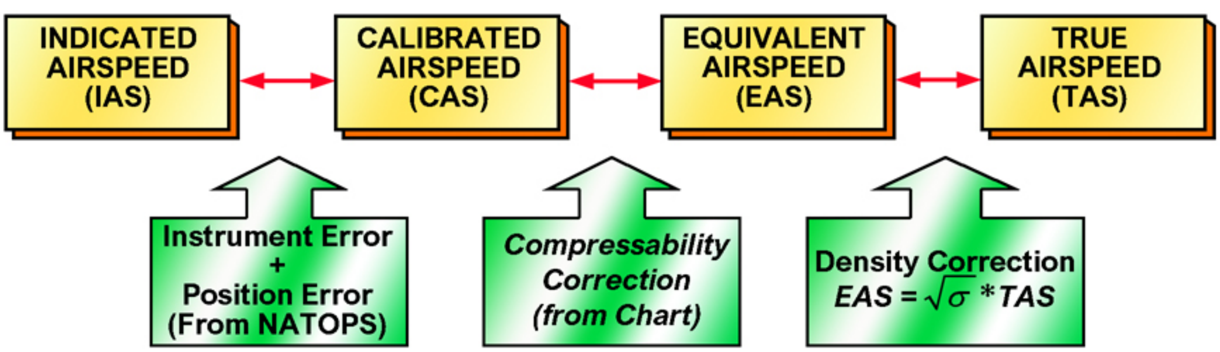


Fig 01. 12 - Airspeed corrections

LIST OF ILLUSTRATIONS

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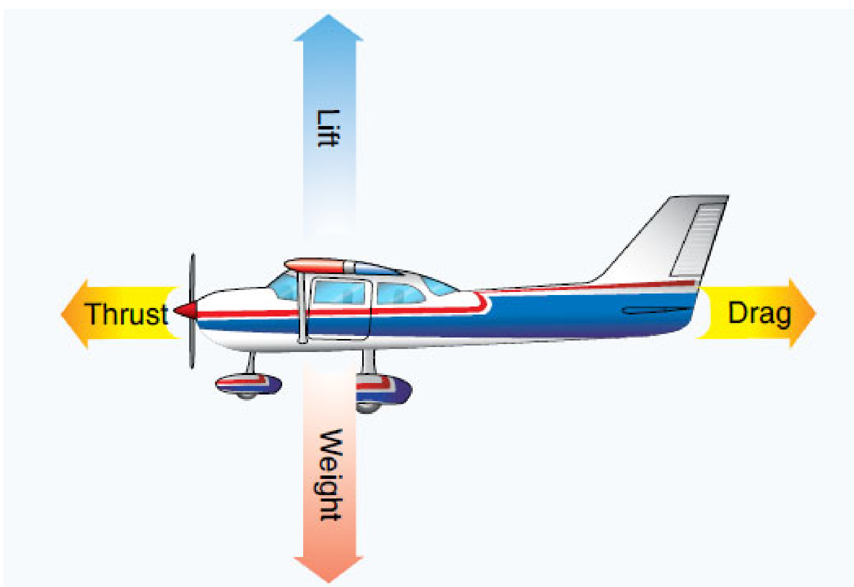

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AERODYNAMICS

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AERODYNAMICS

Air Flow Around A Body

An aeroplane must produce lift to be able to fly before this is discussed; there are some terms and principles belonging to airflow that are to be explained.

Free Stream Flow

Air flow is the movement of air particles in a certain direction. Normally, airflows are not visible. In a storm you can't see the movement of the air particles. You just see the results!

Trees that are blown down, rough seas, broken umbrellas, etc. The air flow caused by an aircraft or a helicopter is not visible either. To be able to observe air flows around objects, the flows have to be made visible first. There are two ways of doing this:

- **By using smoke or vaporised talcum powder.**
- **By using pieces of thread connected to the surface of the object that indicate the flow direction at that particular point.**

Fig 02. 1 - Flow Direction

Laminar Flow and Turbulent Flow

The flow pattern of laminar flows consists of layers. Each air particle follows a certain course and is not influenced by a particle from another layer. There is no difference in velocity between the layers.

Fig 02. 2 - Laminar Flow

Turbulent flows are thet which the velocity and direction change continuously. Some air particles move opposite to others in the air flow. They don't follow their own course but hinder other air particles.

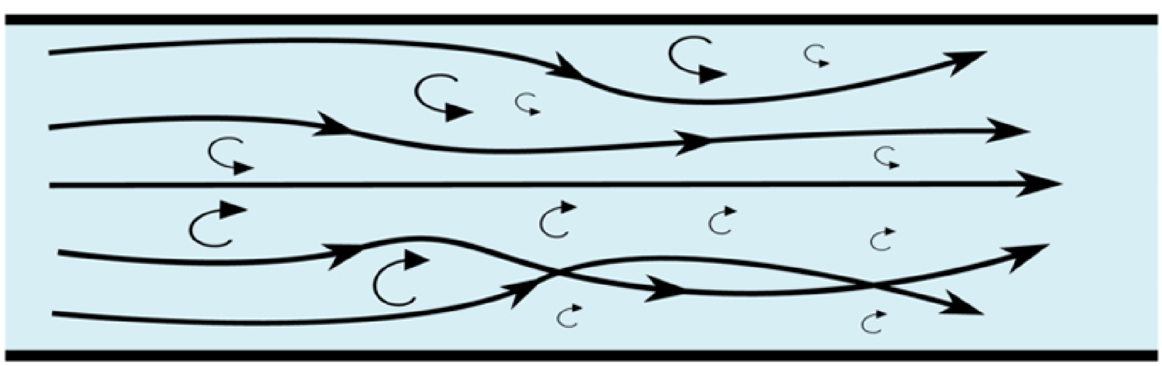

Fig 02. 3 - Turbulent Flow

Critical Velocity

At a certain velocity, laminar flow becomes turbulent flow. This velocity is called the critical velocity. Reaching the critical velocity means that the drag increases suddenly.

If a flat plate is held into slow laminar airflow, parallel to the streamlines, the flow will divide and follow the surface of it.

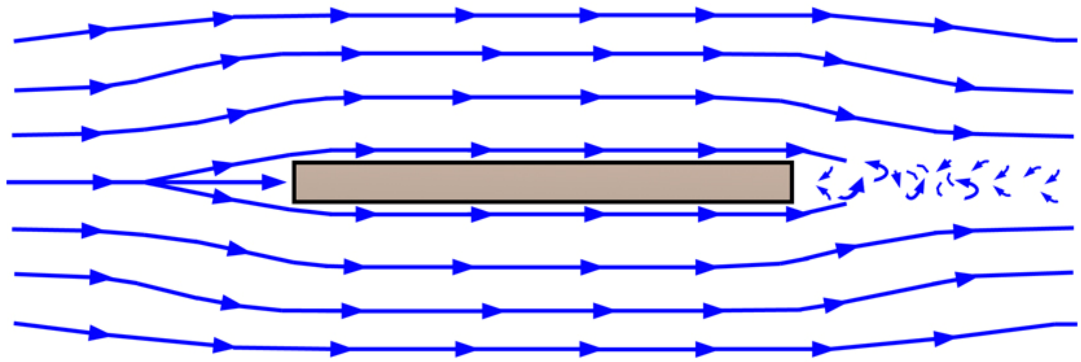

Fig 02. 4 - Air flow around a smooth plate

At the front face part of the air is brought to rest, this area is called stagnation. A closer view to the back face of the plate shows that the air flow turns turbulent in this area.

The patterns of air try to follow the sharp edges, but this is not possible. A curled movement begins which is named vortex. This kind of motion can be observed in every day's life, a good example is water flowing into a sink of a basin or a hurricane.

Vortexes can hide a great amount of energy. Aeroplane also produce vortexes, they cause drag and drag should be prevented as much as possible by a suitable shape of the structure or parts of the structure.

Boundary Layer

Air is viscous and clings to the surface over which it flows. At the surface, the air particles are slowed to near-zero relative velocity. Above the surface the retarding forces lessen progressively, until slightly above the surface the particles have the full velocity of the airstream. The air immediately above the surface is called the boundary layer. It begins at the stagnation area.

The laminar boundary layer increases in thickness with increased distance from the front face edge. Some distance back from the leading edge, the laminar flow begins an oscillatory disturbance which is unstable.

Waviness occurs in the laminar boundary layer which ultimately grows larger and more severe and destroys the smooth laminar flow. Thus, a transition takes place in which the laminar boundary layer decays into a turbulent boundary layer.

The area at which the boundary layer changes from the laminar to turbulent is called the Transition Region.

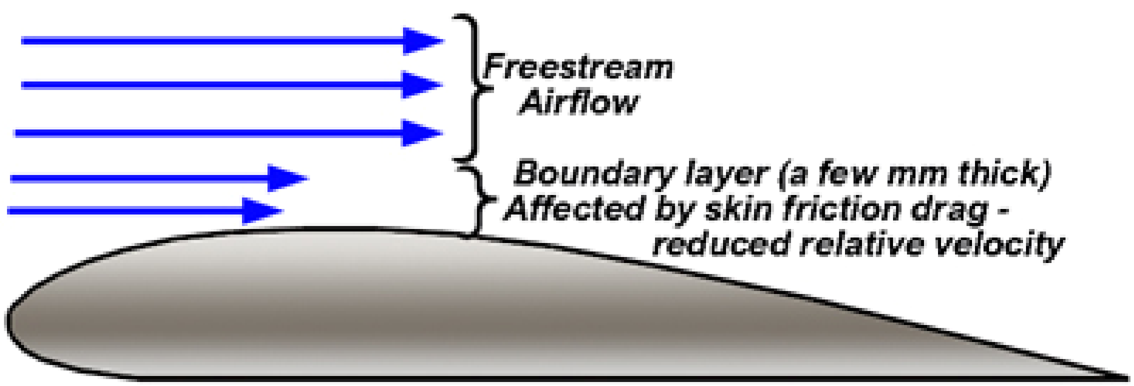

Fig 02. 5 - Development of boundary layer

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Relative Air flow

Because aeroplanes move in the atmosphere and the air also moves (wind), it is necessary to describe the meaning of relative airflow.

Relative airflow is:
The direction of the airflow with respect to an object.

If an object moves forward horizontally the relative airflow moves backward and horizontally. If an object moves forward and upward, the relative airflow moves backward and downward.

Applying this to the movement of an aircraft, the flight path and relative airflow are parallel but travel in opposite directions. Relative airflow may be affected by several factors including wind speed, direction and aircraft altitude.

Technically speaking, the relative airflow is the velocity of the air with respect to a body in it. When the wind blows, the path of the aircraft referred to the ground usually is not the same as the path referred to the moving air.

The aerodynamic forces on the aircraft are always considered as a function of the relative airflow and not of the ground wind.

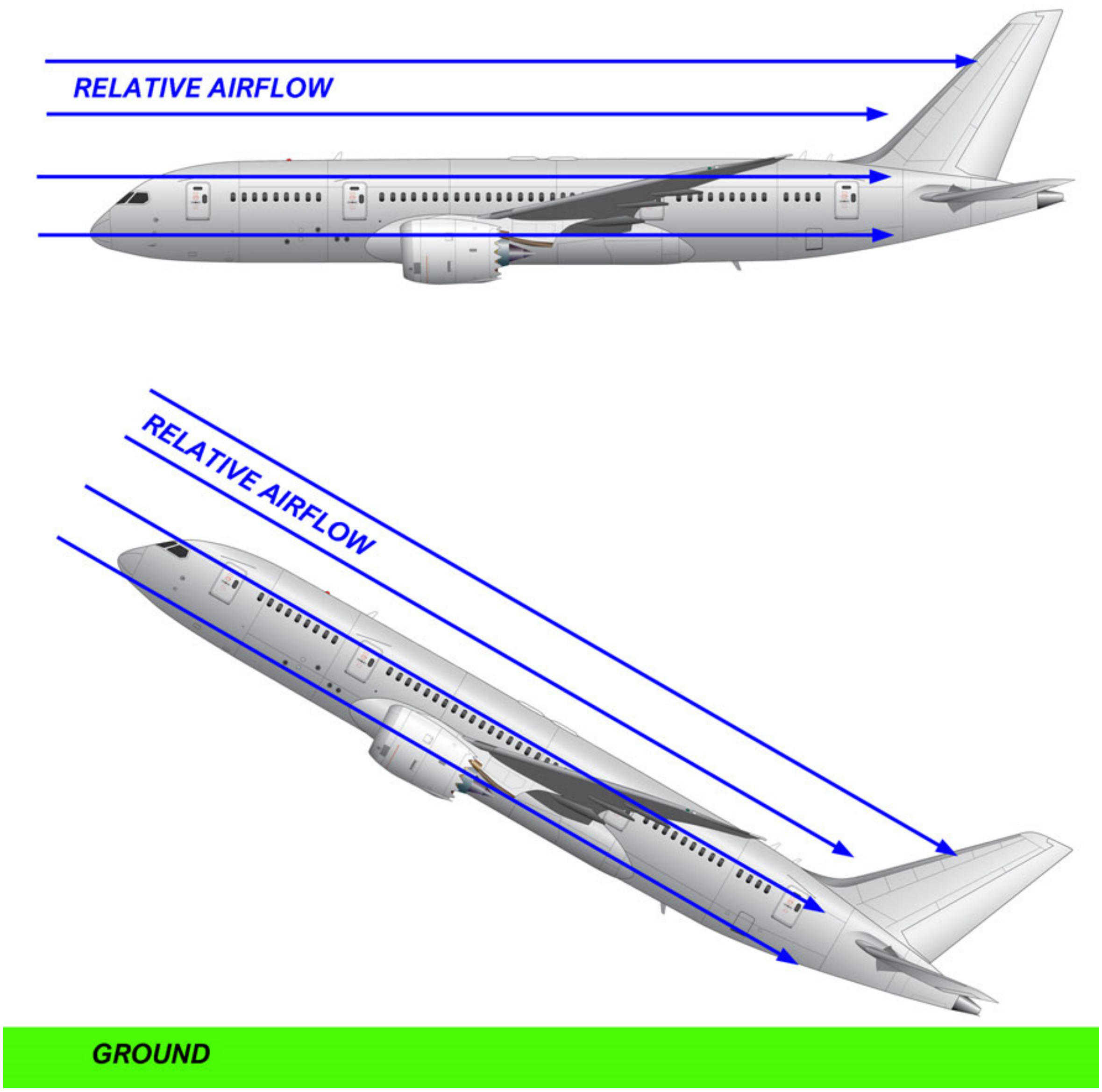


Fig 02. 6 - Relative Airflow

Bernoulli's Principle

The Swiss mathematician and physicist Daniel Bernoulli explains the relationship between potential and kinetic energy in a fluid. This principle is extremely important in helping us understand the way an aeroplane wing or a helicopter rotor produces lift as it passes through the air.

Bernoulli's principle explains that if we neither add nor take away energy from a flow of fluid, any increase in the velocity (kinetic energy) of the flow will cause a corresponding decrease in its pressure (potential energy).

A venturi is a specially shaped tube through which a fluid, either a liquid or a gas, flows. There is a restriction in the tube where the area gets smaller, and then it returns to its original size.

Fluid flows through this tube, and, at position 1, the pressure is the value P1. The speed, or velocity, is shown as V1 if we do not add or take away any energy from the fluid, it must speed up when it reaches the restriction to allow the same volume of fluid to move the same distance through the tube in the same length of time.

When it speeds up, the kinetic energy (velocity) increases, But since no energy was added, the potential energy (pressure) had to decrease. We see this on the gages P2 and V2.

As soon as the tube returns to its original size, at 3, the fluid slows down so the same volume will move the same distance in the same time. When it slows down, the velocity decreases, but the pressure increases to the value it had originally.

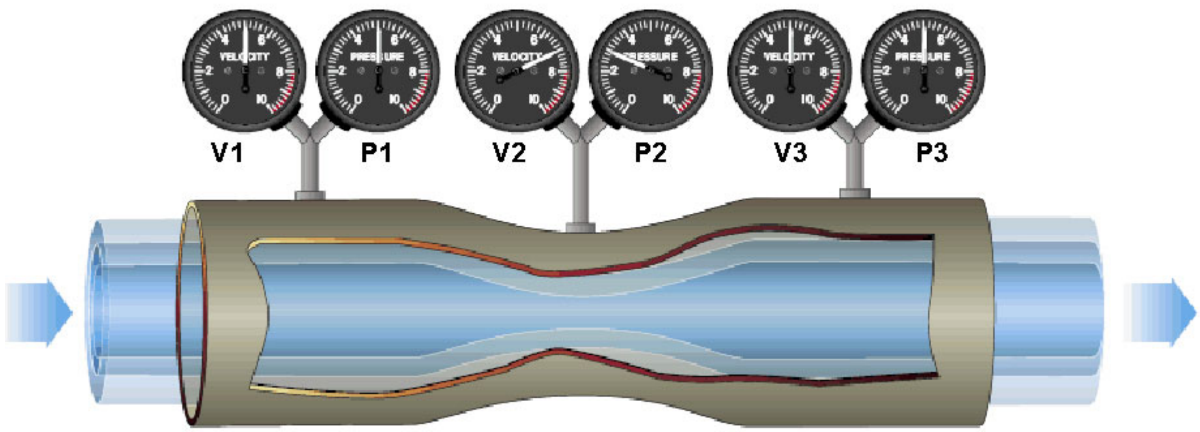


Fig 02. 7 - Venturi Tube

For a flow of air to remain streamlined, the volume passing a given point in a unit of time (the mass flow) must remain constant.

LIFT

Introduction

The curved surface shown below looks like the cross section of an aero plane's wing. It is named airfoil. Prior to studying the generation of lift, it is necessary to have a closer look at the airfoil sections.

Airfoil Sections

Aerodynamic lift depends on the shape of the airfoil section and on the airfoil surface area. The figure bellow shows a typical subsonic airfoil section and some of the more important terms related to its shape.

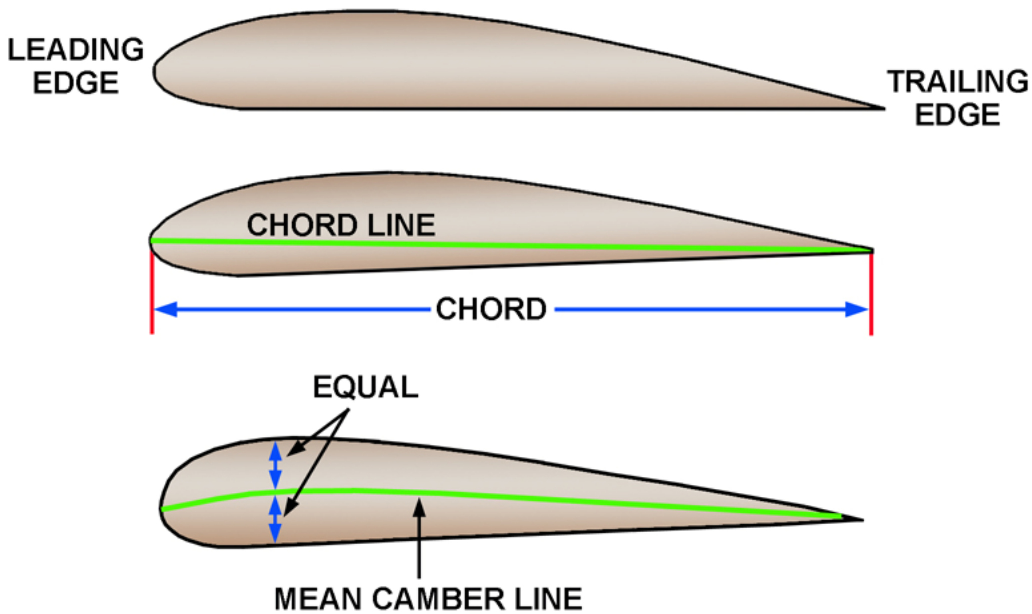


Fig 02. 8 - Airfoil Sections

The mean camber is a line drawn midway between the upper and lower cambers, and its curvature is one of the most important factors in determining the aerodynamic characteristics of the airfoil.

Camber

The maximum camber of a typical low speed airfoil is about 4% of the length of the chord line, and is located about 40% of the chord length behind the leading edge.

The maximum thickness is about 12% of the chord length and is located about 30% of the chord length behind the leading edge.

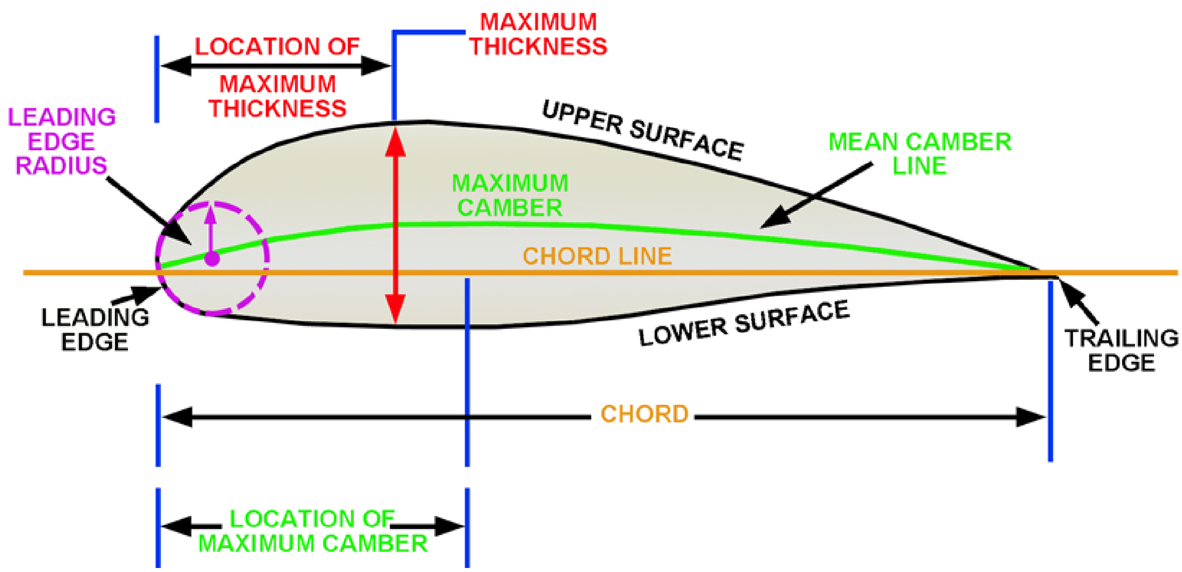


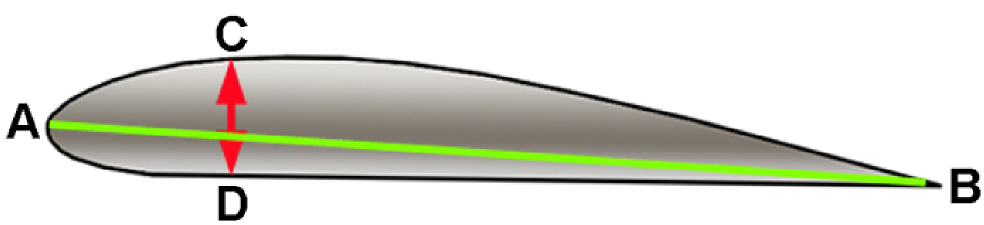
Fig 02. 9 - Airfoil Sections

Fineness Ratio

The ratio of length to breadth is called the Fineness Ratio of a streamlined body.

For best results it should be about 4 to 1, but it really depends on the air speed; the higher the speed, the greater should be the fineness ratio.

Experiments have shown that there is not much variation in the drag for quite a large range of fineness ratios.



$$\text{FINESNESS RATIO} = \frac{AB}{CD}$$

$$\text{THICKNESS/CHORD RATIO} = \frac{CD}{AB}$$

Fig 02. 10 - Airfoil Summary

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History of Airfoils

The earliest airfoils were deeply cambered, and some were not even covered on the bottom. The shape was copied from a bird's wing. The next major step in airfoil development was the Clark-Y airfoil, the standard airfoil section through the 1920s and into the 1930s.

The National Advisory Committee for Aeronautics (NACA), the ancestor of today's NASA, developed much more streamlined airfoils that allowed a smoother flow of air and greater lift with less drag These airfoils included both symmetrical and asymmetrical sections.

When aeroplanes such as the Lockheed Lightning of World War II fame began flying in the transonic range, their subsonic airfoil sections left much to be desired. Shock waves formed, increasing drag and destroying control.

Further study developed the supersonic airfoils, with their maximum thick- ness about midway back and their equally sharp leading and trailing edges.

The super critical airfoil evolved next, with its blunter leading edge and flatter upper surface. This airfoil section reduces the velocity of the air over the upper surface and delays the extreme drag rise that occurs as the airfoil approaches the speed of sound.

The NASA-developed GAW- 1 and GAW-2 low-speed airfoils have the same downward cusp at the trailing edge that the super critical airfoil has. These airfoils were developed for general aviation aircraft, and they give a wing a higher LID ratio by increasing the lift and decreasing the drag.

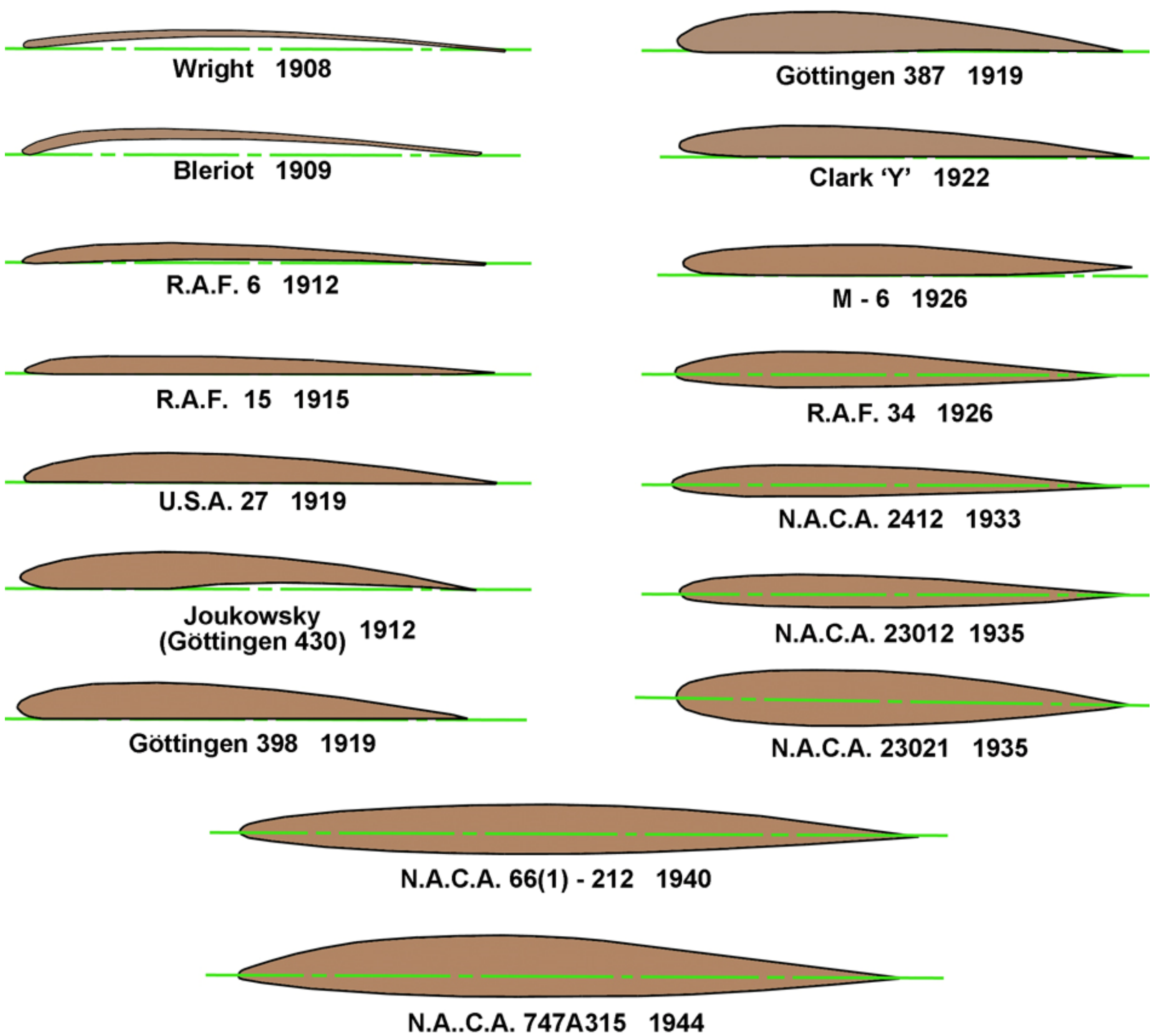

Fig 02. 11 - Airfoil Evolution

AERODYNAMIC FORCES

Lift

When the angle between the airfoil and airflow (angle of attack) increases, the asymmetrical flow causes a pressure differences between the upper and lower surfaces of the wing. This creates upward and down-ward forces.

The air particles go faster over the upper surface of the wing. Their speed depends on the angle of the wing and the airflow in relation to each other. Because of the increase in speed, the static pressure decreases on the top of the wing (Bernoulli's Law).

A pressure difference occurs between the upper and lower surfaces of the wing and the airflow, just before reaches the leading edge, is sucked into this low-pressure area. This is called upwash.

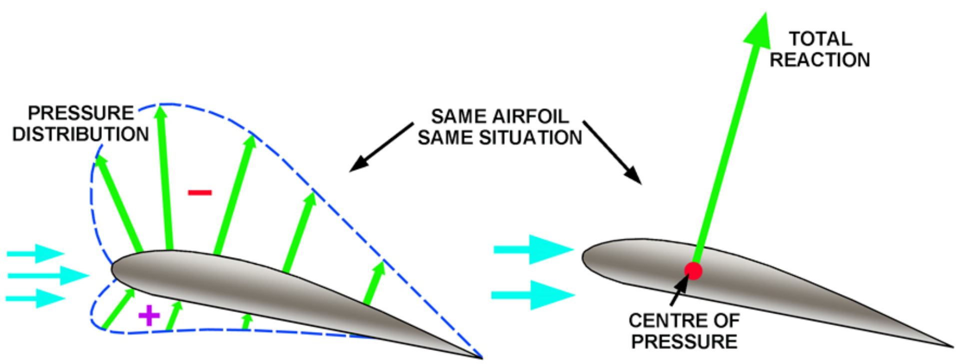

Fig 02. 12 - Development of Aerodynamic Lift

The amount of lift is determined by, among others, the shape of the wing. In the case of a flat wing shape (parallel to the direction of flow) an equal number of air particles flow along the upper and along the lower surfaces of the wing. The drag is minimal and the pressure difference between top and bottom are nonexistent.

In the case of a cambered wing upper surface, the air particles along the top must travel a greater distance than the air particles moving along the bottom surface of the wing.

According to the rule of continuity, the number of in flowing air particles per unit of time is equal to the number of out flowing air particles. This means that the particles along the top of the wing must move faster than the air particles along the bottom of the wing. Under pressure is created.

It also means that the pressure on the underside is greater than on the topside. Taken together, they cause an upward force. The heavier the aeroplane is, the greater the required lift must be.

The air above the surface will be pulled down into this low-pressure area and, as a result, air is forced down as the airfoil moves through it. In order to support an aircraft, the weight of the air forced down must equal the weight of the aircraft.

The amount of lift produced can be increased by increasing the angle of attack, and thus the downwash angle. But a maximum angle will be reached beyond which this is no longer true. Above this critical angle, the air will break away from the upper surface, and the airfoil will stall.

Also, the faster the airfoil is moved through the air, the greater will be the amount of air deflected and the amount of lift produced.

- **LIFT**
  - **Is the component of the total reaction at right angles, or perpendicular to the relative airflow.**
- **DRAG**
  - **Is the component of the total reaction parallel to the relative airflow and opposing motion.**

- **Relative Airflow**
  - **This refers to the motion between a body and the remote airflow, i.e. the airflow far enough away from the body nou to be disturbed by it.**
- **Alnge of Attack (AOA)**
  - **The angle between the Chord Line and the Relative Airflow.**

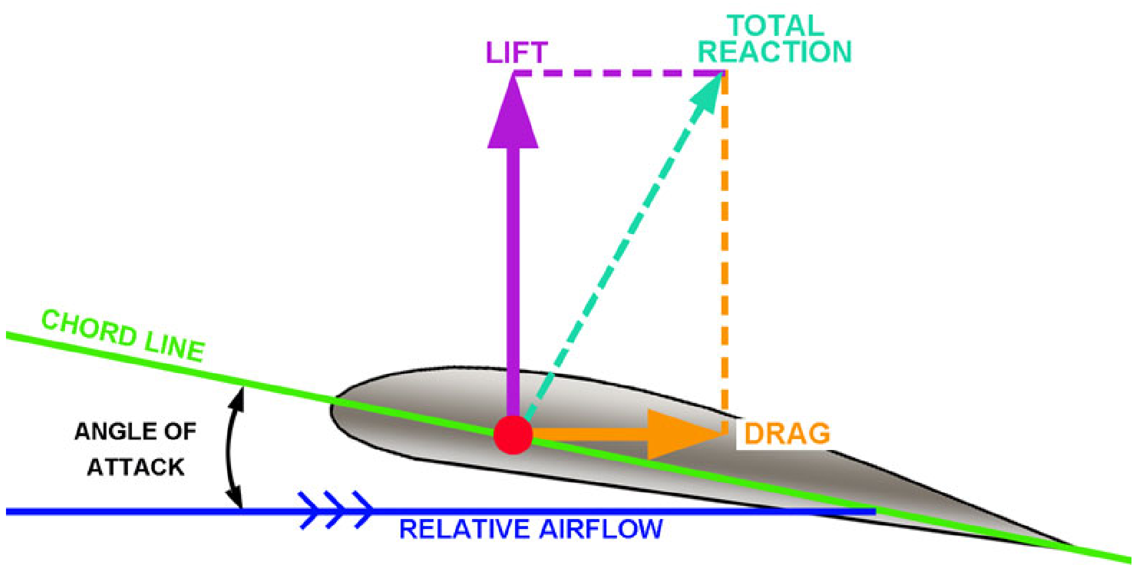

Fig 02. 13 - Development of Aerodynamic Lift

Angle of Attack

Altering the angle that the aerofoil meets the oncoming air can control the amount of lift produced. This is called the 'Angle of Attack'.

The aerofoil is most efficient when set at an angle of approximately 4° to the oncoming airflow.

NOTE:

DO NOT confuse Angle Of Attack with Angle Of Incidence.

This is the angle at which the wing is fixed to the airframe relative to the longitudinal axis.

The angle of incidence is fixed, but the angle of attack changes in flight.

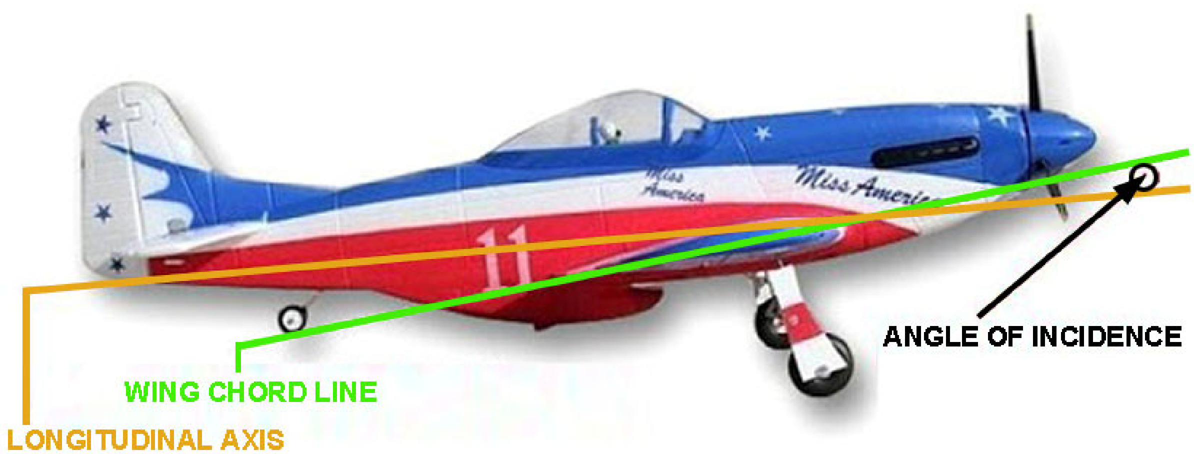

Fig 02. 14 - Angle of Incidence

Lift Formula

$$LIFT = C_L \, {}^1\!/_2 \, \rho V^2 S$$

Experimentally it can be shown that the total reaction and therefore the lift , depends on:

- **Wing shape**
- **Angle of attack**
- **Air density (rho)**
- **Freestream velocity(V)**
- **Wing surface area(S)**

Drag

Drag is the aeronautical term for air resistance experienced by the aeroplane as it moves relative to the air. It acts in the opposite direction to the motion through the air, i.e. it opposes the motion and acts parallel to, and in the same direction as, the relative airflow.

Low drag requires only low thrust to balance it.

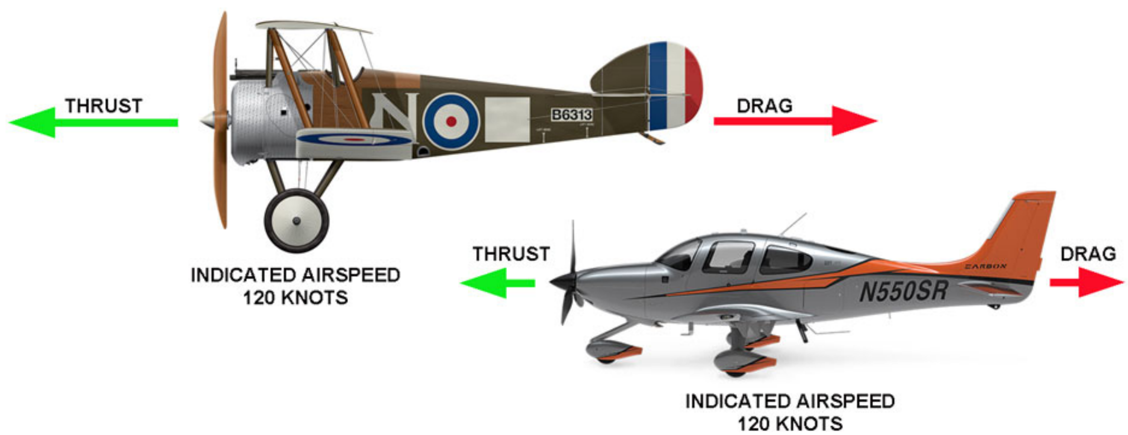

Fig 02. 15 - Drag

The total drag on an aircraft is the sum of all of the types of drag. Drag can be classified into two main headings, these are:

- **Induced Drag**
  - **Drag due to lift.**
- **Parasite Drag**
  - **Drag due to the viscosity of the air.**

Induced Drag

This is inseparable from lift and any time that lift is produced induced drag will be present.

It has been shown that when an airfoil produces lift there is a positive pressure beneath the wing. This pressure moves outwards toward the lower pressure of the atmosphere.

On the top surface there is a negative pressure which moves inwards away from the higher pressure of the atmosphere.

As the two airflow's mix together at the trailing edge, there is a twisting motion imparted because of the opposite relative flow directions. This forms a series of small vortices which travel in a downward direction.

The Vortices become progressively larger from wing root to wing tip. At the wing tip high pressure air from beneath the wing spills over into the low pressure on top of the wing, this causes a large trailing vortex.

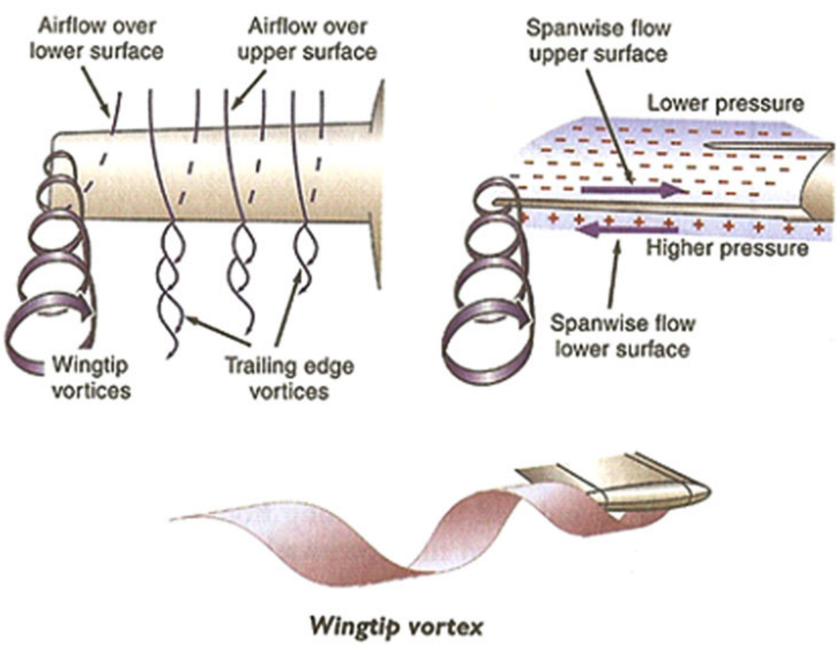

Fig 02. 16 - Wing Tip Vortices

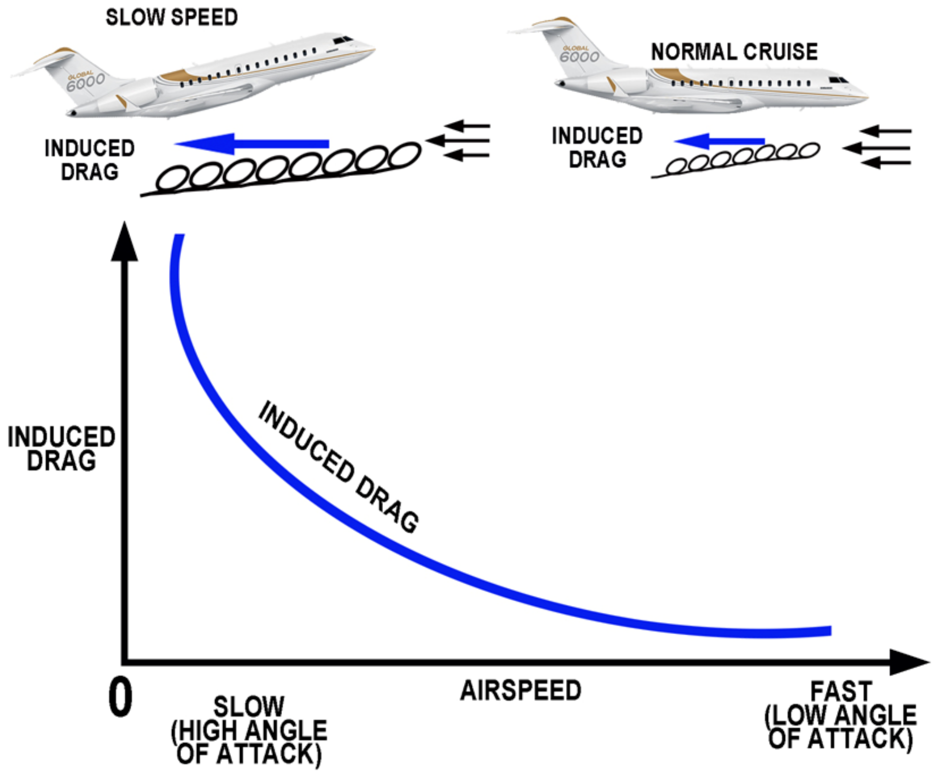

Fig 02. 17 - Induced Drag

Parasite Drag

Parasite Drag can be broken down into three types

- **Form Drag**
- **Interference Drag**
- **Skin Friction**

Any part of the aircraft which is presented to the relative airflow will produce drag due to the resistance of the air flowing past it.

Any part of the aircraft which does not contribute to lift will produce Parasite Drag. Typically landing gear, antennas, and the fuselage all produce Parasite Drag.

Form Drag

This is the portion of the resistance which is due to the fact that when a viscous fluid flows past a solid object, vortices are formed, and we no longer get a smooth streamline flow. The extreme example of this type of resistance is a flat plate placed at right angles to the airflow. The resistance is very large and is almost entirely due to the formation of vortices, the skin friction being negligible in comparison.

Experiments show that not only is the pressure in front of the plate greater than the atmosphere pressure, but that the pressure behind is less than that of the atmosphere, causing a kind of "sucking" effect on the plate.

It is essential that form drag should be reduced to a minimum in all those parts of the aircraft which are exposed to the air. This can be done by so shaping them that the flow of air past them is as smooth as possible, and much experimental work has been carried out with this in view.

The results show the enormous advantage to be gained by the streamlining of all exposed parts, in fact, the figures obtained are so remarkable that they are difficult to believe without a practical demonstration.

At a conservative estimate it can be said that a round tube has not much more than half the resistance of a flat plate, while if the tube is converted into the best possi• ble streamline shape the resistance will be only one-tenth that of the round tube or one-twentieth that on the flat plate.

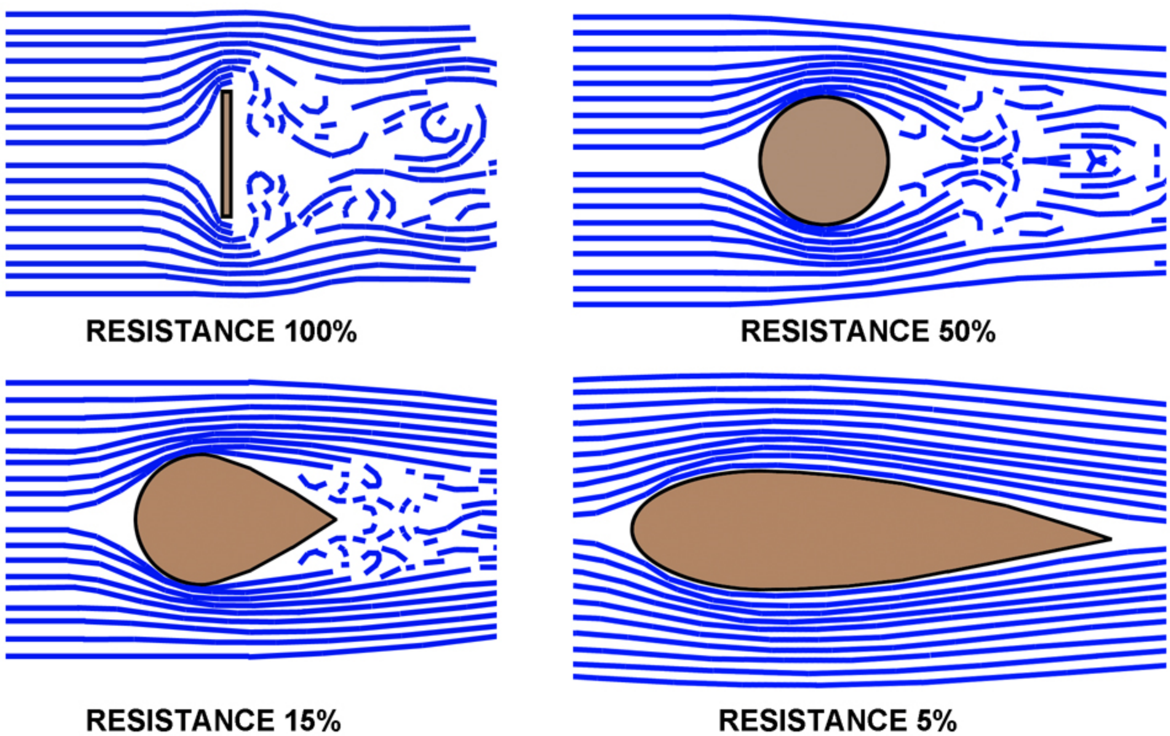

Fig 02. 18 - The Effect of Streamlining

Interference Drag

Samples of areas affected by interference drag are; wing to fuselage attachment, or engine nacelles to wing connection. The interference drag is difficult to appraise. To reduce this drag it is necessary to design the transition surfaces between two bodies without sharp edges, e.g. the connection between wing and fuselage is designed as a fillet.

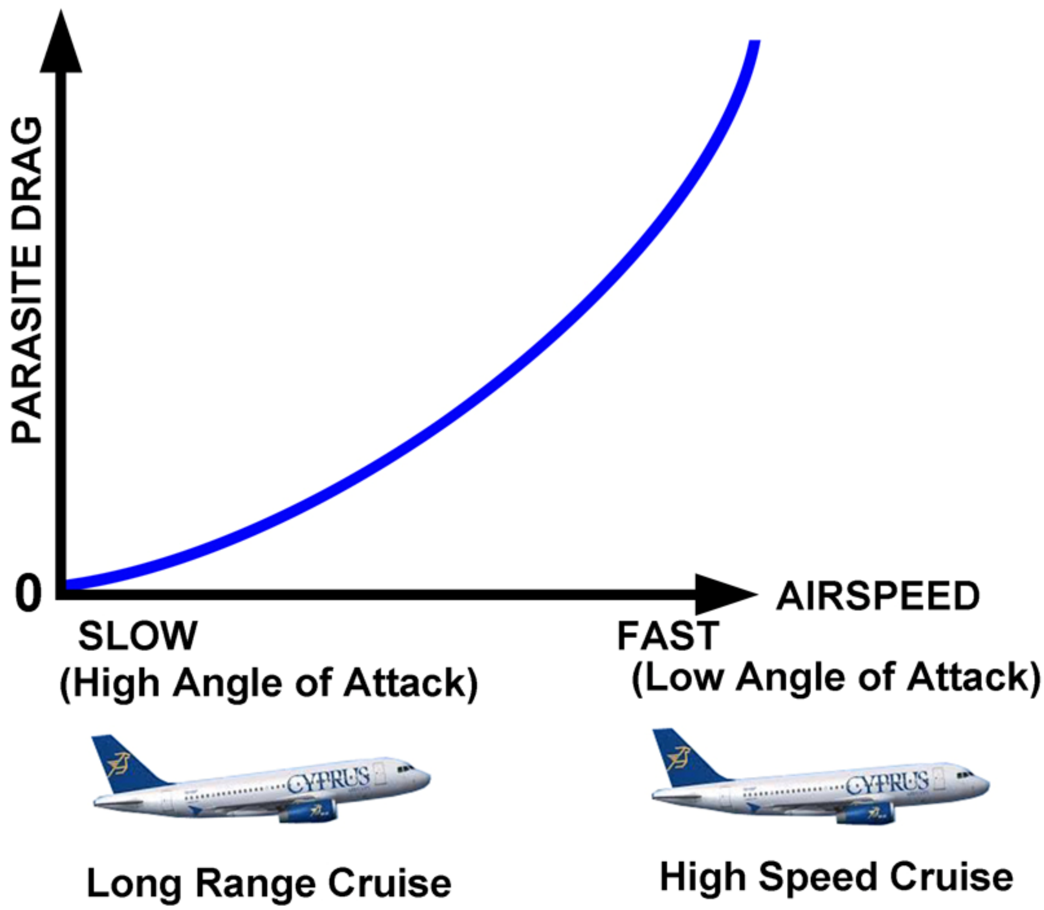

Fig 02. 19 - Interference Drag

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Total Drag

To find the total drag of an aircraft, we must add the parasite and the induced drag, and in Figure 31 we see that we have a minimum drag at the airspeed at which the induced and parasite drag are the same. If we fly slower, we need a larger angle of attack, and the induced drag goes up. If we fly faster, the parasite drag predominates and increases the total drag.

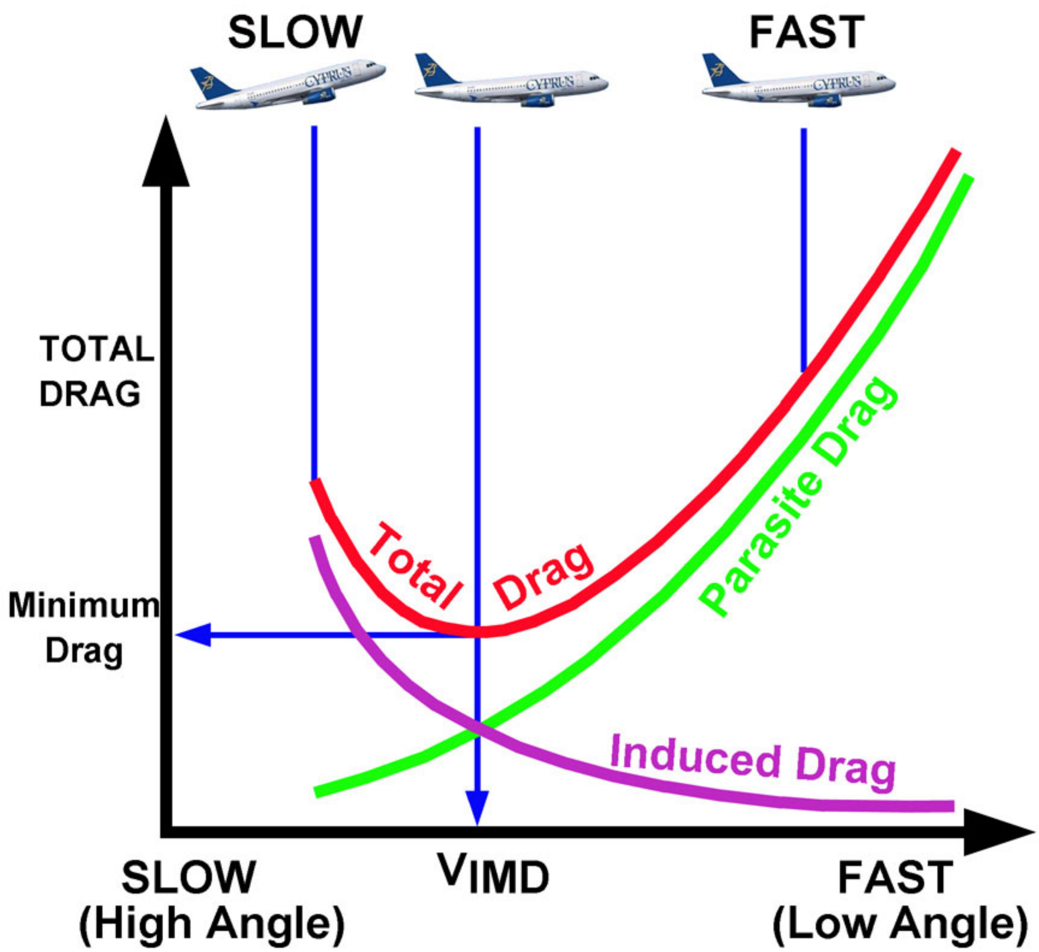

Fig 02. 20 - Total Drag

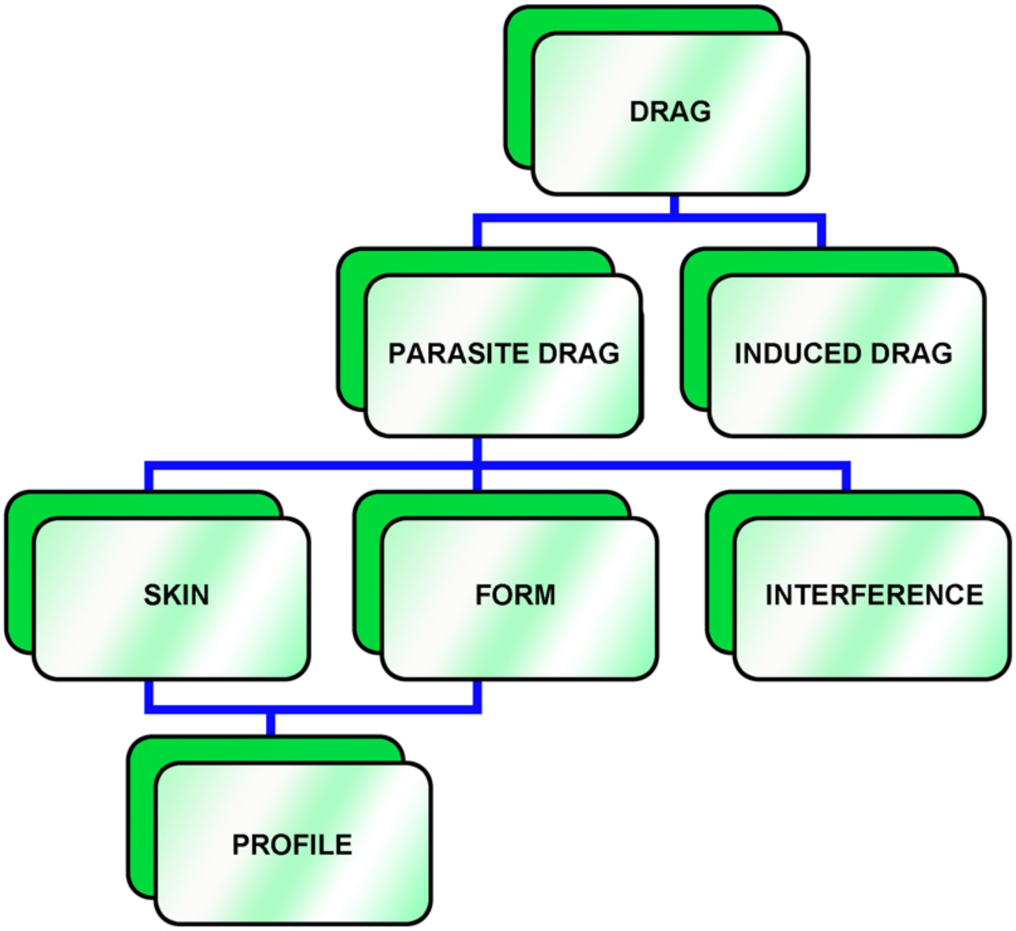

Fig 02. 21 - Different Types of Drag

Induced Drag Calculation

$$DRAG = C_D \, {}^{1}/_{2} \, \rho V^2 S$$

Cd = drag coefficient (represents shape and angle of attack

ρ = air density

S = wing area (m2)

V= velocity

Development of Lift and Drag

Five factors affect aerodynamic lift and drag:

- **Shape of the Airfoil Section**
- **Area of the Airfoil**
- **Air Density**
- **Speed of the Air relative to the airfoil surface**
- **Angle between the Airfoil and the Relative Airflow (the angle of attack)**

Notice that two of these factors relate to the airfoil, two to the air, and one to the relationship between the two.The direction of the lift produced by the wing is always perpendicular to the direction of the relative airflow.

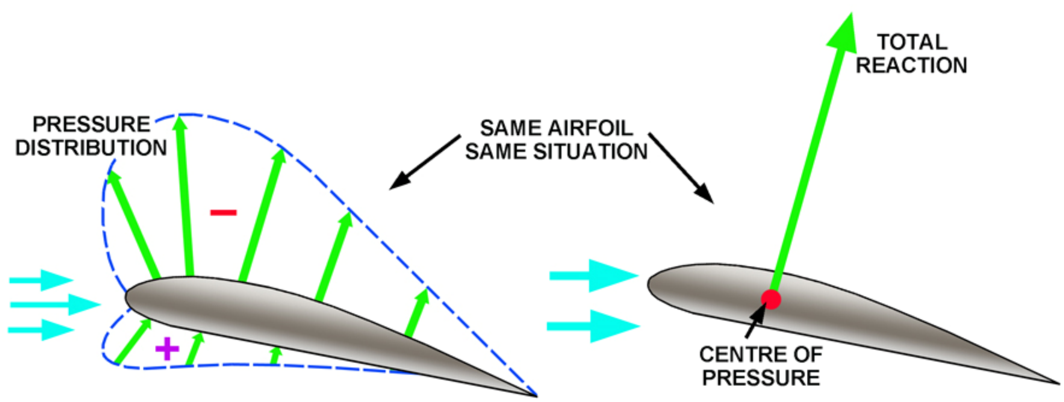

Fig 02. 22 - Centre of pressure

The center of pressure is the point on the chord line of an airfoil at which all of the aerodynamic forces are concentrated. The lift vector acts from the center of pressure in a direction that is perpendicular to the relative airflow, and the drag vector acts from this same point in a direction parallel to the relative airflow.

The center of pressure of a subsonic airfoil is typically located somewhere around 30% to 40% of the chord line back from the leading edge. On an airfoil, the center of pressure moves forward as the angle of attack increases and backward as it decreases.

The total reaction changes size and the centre of pressure moves as the angle of attack changes. At the 'Stall' position the wing loses lift, the airflow becomes turbulant and the centre of pressure moves rearwards.

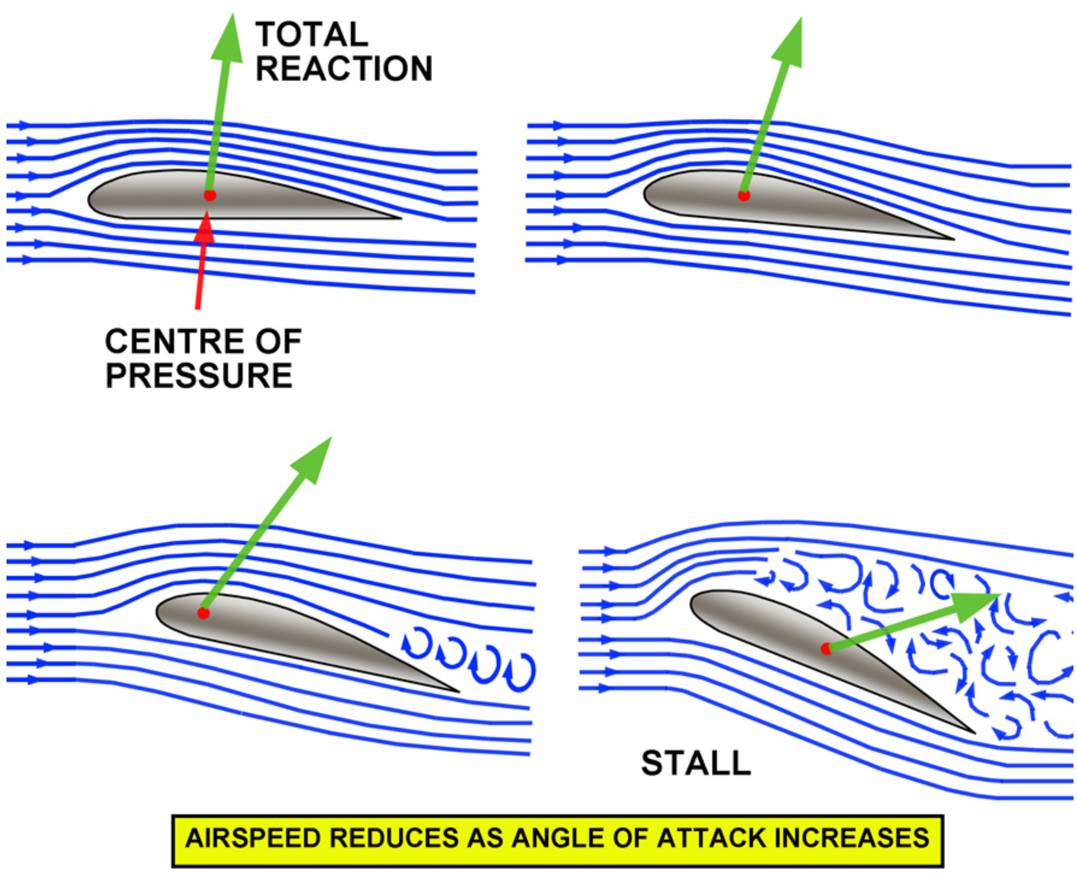

Fig 02. 23 - Centre of Pressure Movement

THE WING

Introduction

It is not only the airfoil shape, which determines the amount of lift produced from an aeroplane's wing. There are many other wing dimensions which contribute to the produced amount of Lift. The most important ones are described in this section.

At first some terms belonging to wing are described.

Wing Sections

- **Span (b)**
  - **Is the maximum distance from tip to tip.**
- **Area (S)**
  - **Is the surface of the planview of the apparent wing, including the parts covered from the fuselage, engines and others.**

Aspect Ratio

Early in aerodynamic history, it was discovered that the aerodynamic characteristics of an aircraft depended upon the ratio of the wing span to the wing chord. For a given amount of lift, the chord required reduces as the wing span is increased.

Thus, wings designed to minimise trailing-vortex (induced) drag, have a long span, with a small chord: in other words, the aspect ratio is high. For a glider or sail plane, the aspect ratio may be as high as 40.

Commercial aircraft have aspect ratios between five and eight, while jet fighters that operate at transonic and low supersonic speeds have wings of aspect ratios near 3.5.

Effects of Aspect Ratio

The effect of increasing aspect ratio is principally to reduce induced drag for any given amount of lift. The effect of higher aspect ratio is to increase the range of an aircraft because there is less drag to overcome.

Subject to certain limitations, the most efficient airfoil has the greatest aspect ratio. A true cantilever wing has no external bracing. If it has a high aspect ratio, it is long and narrow. It is obvious that when the length is increased too much, the wing will droop from its own weight

Furthermore, the root section (where it joints the fuselage) must be of heavy construction, and this added weight may in itself offset some or all of the gain in lift.

When a wing reaches a certain length, it requires external bracing, which adds to the weight and causes increased drag.

$$\textit{Aspect Ratio} = \frac{SPAN}{CHORD}$$

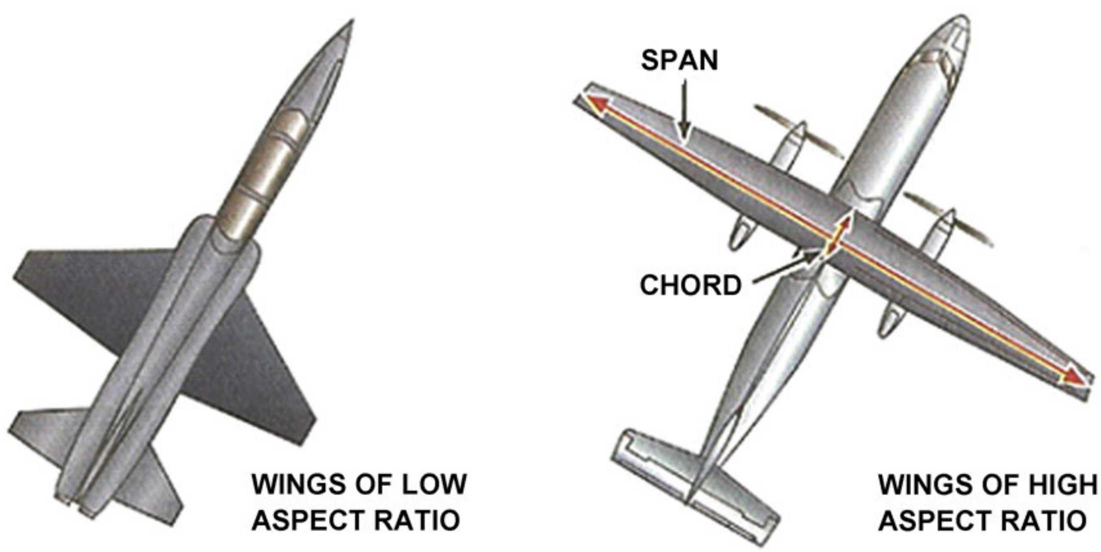

Fig 02. 24 - Aspect Ratio

Mean Aerodynamic chord (MAC)

The distance between the leading and trailing edge of the wing, measured parallel to the normal airflow over the wing, is known as the chord. If the leading edge and trailing edge are parallel, the chord of the wing is constant along the wing's length.

Most commercial transport airplanes have wings that are both tapered and swept with the result that the width of the wing changes along its entire length. The width of the wing is greatest where it meets the fuselage at the wing root and progressively decreases toward the tip.

As a consequence, the chord also changes along the span of the wing. The average length of the chord is known as the mean aerodynamic chord (MAC).

In large aircraft, centre of gravity limitations and the actual centre of gravity are often expressed in terms of percent MAC.

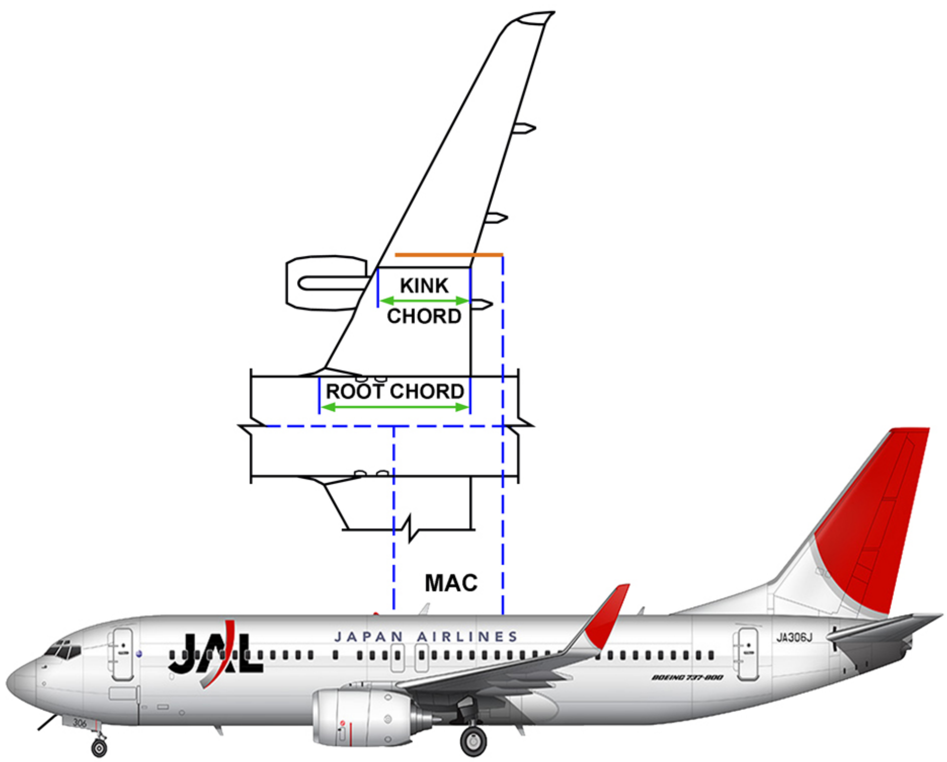

Fig 02. 25 - Mean Aerodynamic chord

Washout

Many aircraft are designed with a greater angle of incidence at the root of the wing than at the tip; this characteristic of a wing is called washout

The purpose of washout is to improve the stability of the aircraft as it approaches a stall condition. The section of the wing near the fuselage will stall before the outer section, thus enabling the pilot to maintain good control and reducing the tendency of the aircraft to "fall off' on one wing.

If a wing is designed so that the angle of incidence is greater at the tip than at the root, the characteristic is called washin.

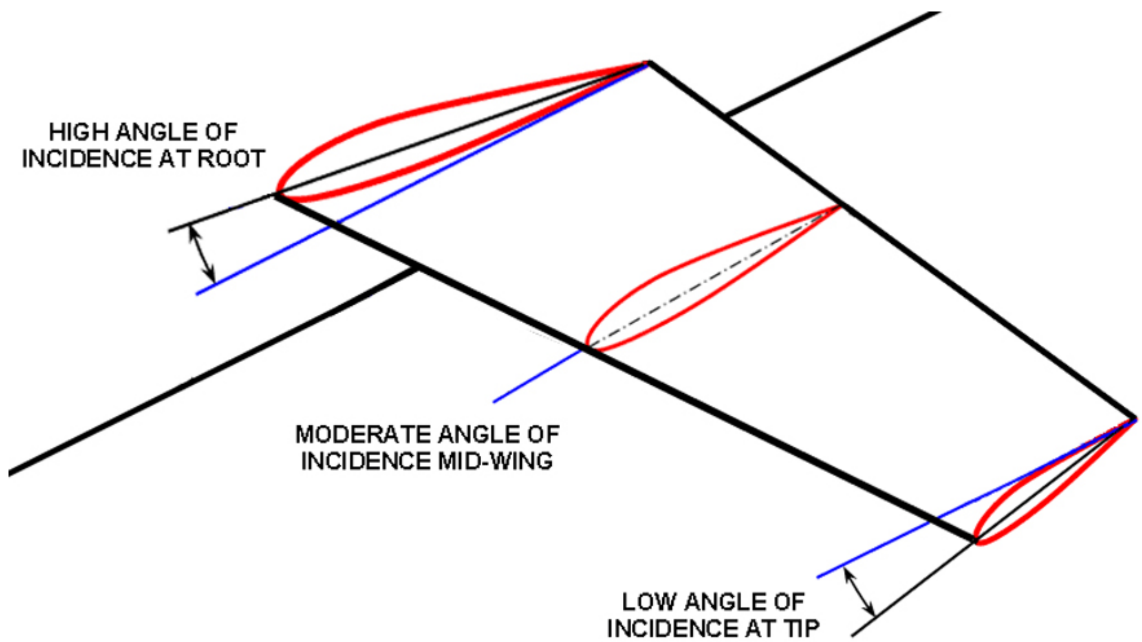

Fig 02. 26 - Washout

AIRFOIL CONTAMINATION

Ice Accumulation

Rain, snow, and ice are transportation's ancient enemies. Flying has added a new dimension, particularly with respect to ice. Under certain atmospheric conditions, ice can build rapidly on airfoils and air inlets.

The accumulation of ice on the leading edge of wings can significantly alter the geometry of the wing and cause drag penalty and performance degradation (and in the worst case, safety can be affected. The two types of ice encountered during flight are "rime" and "glaze".

Rime ice forms a rough surface on the aircraft leading edges. It is rough because the temperature of the air is very low and freezes the water before it has time to spread.

Glaze ice (clear ice) forms a smooth, thick coating over the leading edges of the aircraft. When the temperature is just slightly below freezing, the water has more time to flow before it freezes.

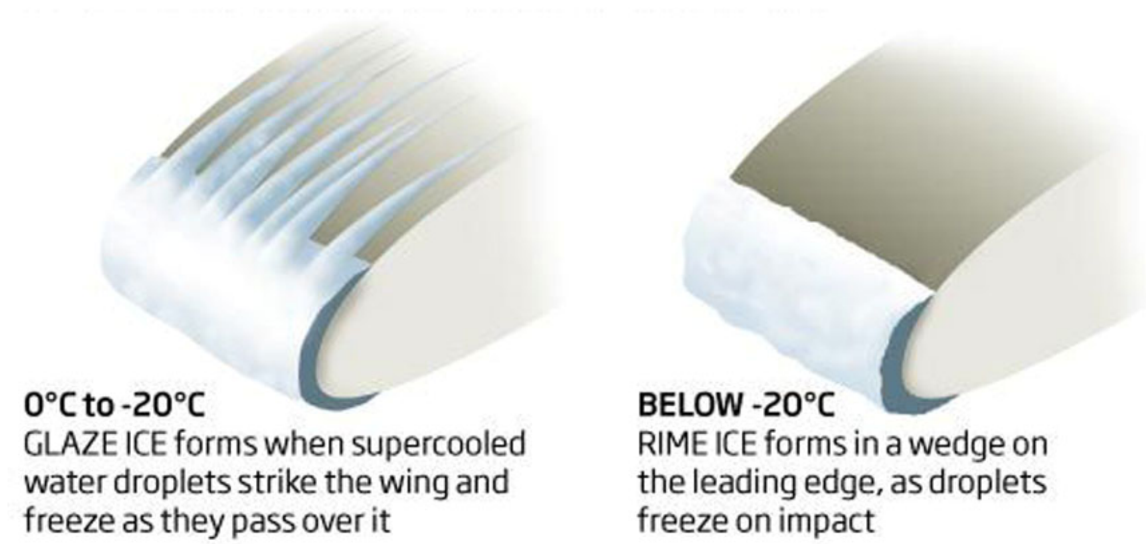

Fig 02. 27 - Ice Formations

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Contamination Effects

Ice on an aircraft affects its performance and efficiency in many ways. Ice buildup increases drag and reduces lift. It causes destructive vibration, and hampers true instrument readings.

Control surfaces become unbalanced or frozen. Fixed slots are filled and movable slots jammed. Radio reception is hampered and engine performance is affected.

Ice, snow, rime formation or deteriorations of other kind having a thickness and surface roughness of a medium sandpaper on the leading edge and upper surface of the wing can reduce wing lift by around 30% and increase drag by 40%.

It is important to consider that other kind of contaminations or deteriorations also affect wing Lift and drag in the same way like explained for ice, rime and snow. Some examples of contamination and deterioration are:

- **Misrigging of Control Surfaces.**
- **Absence of Seals on Movable Sections.**
- **Dents on Surfaces (bird strikes, accidental ground damages).**
- **External Repair (doubler).**
- **Paint Peeling.**
- **Lack of Cleanless.**

The first 20% of the wing chord is particularly sensitive, because disturbancy of air flow passing it can cause separation resulting in early stall.

These changes in Lift and drag significantly increase stall speed, reduce controlability and alter flight characteristics.

Fig 02. 28 - Wing and Aircraft Ice Contamination

Stalling

In an earlier note it was stated that the nature of the boundary layer determined the stalling characteristics of a wing. In particular the phenomenon of boundary layer separation is extremely important. This note will discuss what happens when a wing stalls.

Ideally the airflow around an aerofoil is streamline. In real life the streamline flow breaks away (or separates) at some point from the aerofoil surface and become turbulent. At low angles of attack this separation point is towards the rear of the wing and the turbulence is not significant.

At higher angles of attack the separation point moves forward. As the angle of attack is increased, a critical angle is reached beyond which the separation point will suddenly move well forward, causing a large increase in the turbulence over the wing.

The formation of low static pressure on the upper surface of a wing (the main contributor to the generation of the lift force) is reduced by the breakdown of streamline flow. Turbulence flow does not encourage the formation of low static pressure areas.

The lifting ability of a wing (coefficient of liftC_L) decreases markedly beyond this critical angle of attack as a result of the breakdown of streamline flow.

The significant breakdown of streamline flow into turbulence over a wing is called stalling of the aerofoil. This critical angle or stalling angle of attack is where C_L reaches its maximum value and beyond which C_L decreases markedly.

Beyond the stalling angle, the center of pressure (which has been gradually moving forward as angle of attack increases) suddenly moves rearwards and there is also a rapid increase in drag.

NOTE:

A stall occurs at the critical angle of attack only!

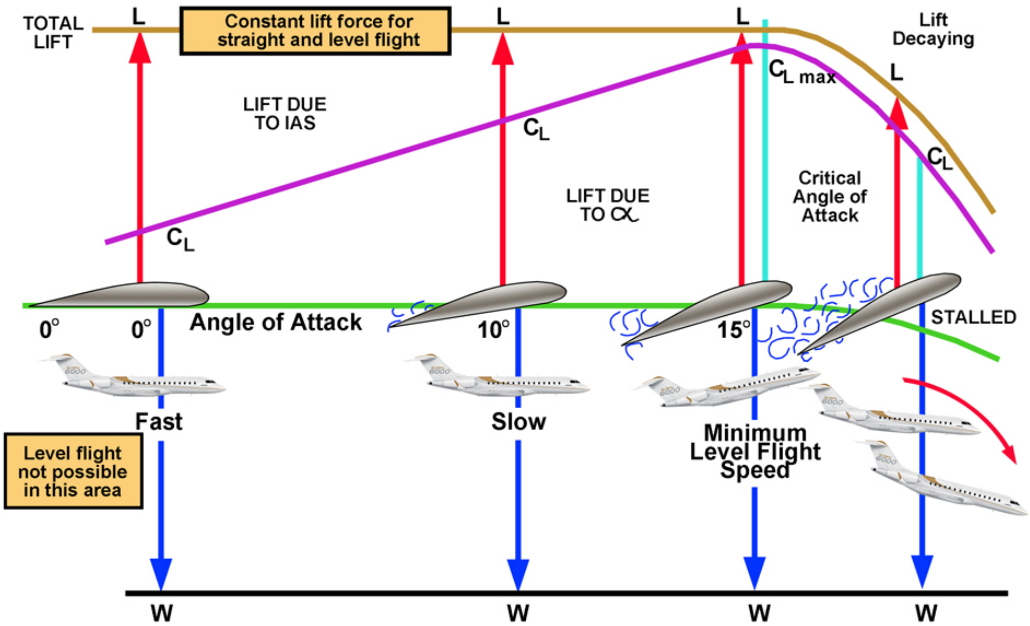

Fig 02. 29 - Stalling

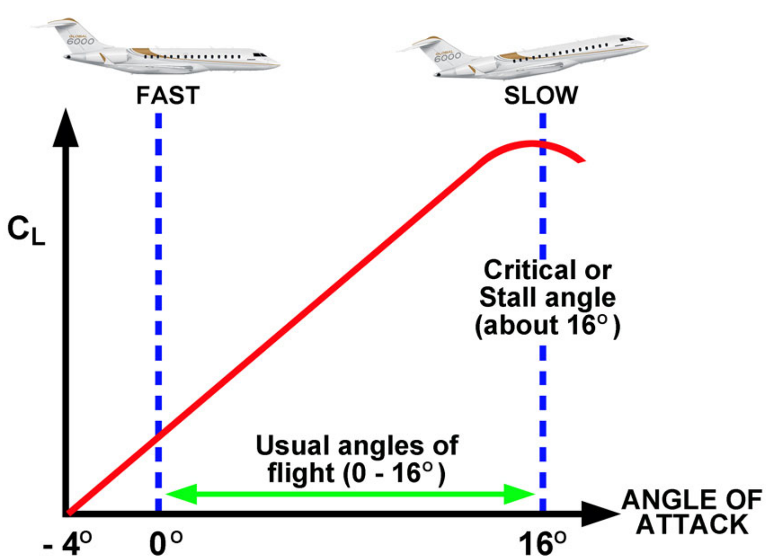

Fig 02. 30 - Coefficient of Lift

LIST OF ILLUSTRATIONS

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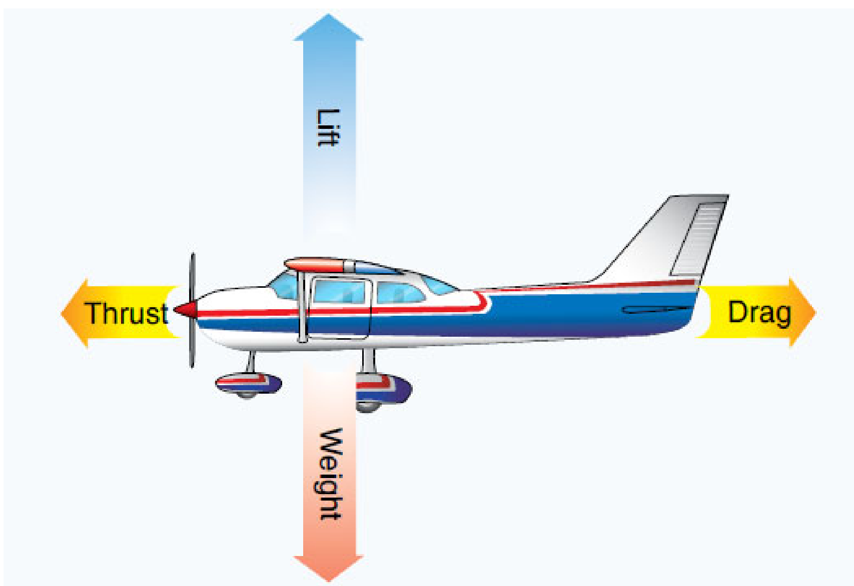

SECTION 08-03 A1/B1/B2

THEORY OF FLIGHT

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THEORY OF FLIGHT

Forces Acting On The Aircraft In Flight

When in flight, there are certain favourable forces and other unfavourable forces acting on the aircraft. Among the aerodynamic forces acting on the aircraft during flight, four are considered to be basic because they act upon the aircraft during all manoeuvres.These basic forces are:

- **Lift**
- **Thrust**
- **Weight**
- **Drag**

Lift

In a classic configuration, nearly all lift is generated by the wings. Lift is the force which supports the aircraft in flight. It acts at right angles to the relative airflow and the chord line. In steady constant velocity flight lift exactly equals weight.

Weight

Gravity is the downward force which tends to draw all bodies vertically towards the center of the earth. The aircraft's Centre of Gravity (CG) is the point on the aircraft at which all weight is considered to be concentrated. It is the point of balance.

The center of gravity is located along the centerline of the aircraft and forward near the centre of lift of the wing. The location of the center of gravity depends upon the location and weight of the load placed in the aircraft.

Thrust

Is the force produced by the power plant. It acts in a forward direction and is used to overcome the force of drag.

Drag

As we have already mentioned ‘Drag’ is the aeronautical term for air resistance experienced by the aeroplane as it moves relative to the air. It acts in the opposite direction to the motion through the air, i.e. it opposes the motion and acts parallel to, and in the same direction as, the relative airflow.

Balancing the four Forces

In level constant velocity flight, Lift equals Weight and Thrust equals Drag.

The four forces do not necessarily act at the same point and move around as the lift, weight, thrust and drag change. The forces have to be arranged in a manner so that the moments produced by the forces, balance each other out during straight and level flight.

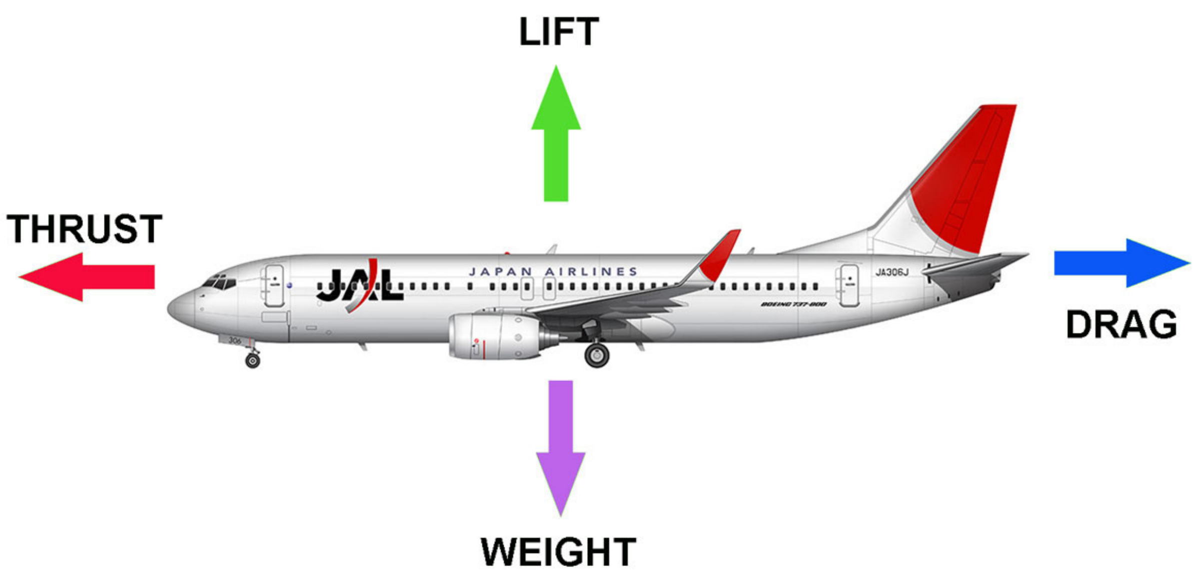

Fig 03. 1 - The Four Basic Forces

Pitching Moments

- **If the Centre of Pressure (CP) is acting behind the CG this will give the aircraft a nose down pitching moment.**
- **If the CP is acting forward of the CG this will give the aircraft a nose up pitching moment.**
- **If the thrust line is below the CG this will give the aircraft a nose up pitching moment.**
- **If the thrust line is above the CG this will give the aircraft a nose down pitching moment.**

Ideally, the pitching moments set up from Lift - Weight and Thrust - Drag, should cancel each other out in straight and level flight so as to have no residual moments, tending to pitch the aircraft.

It is normal to design an aircraft so that the center of pressure lies slightly behind the CG with the thrust line below the CG. Then the arrangement Lift - Weight will give a nose down moment while Thrust - Drag will give a nose up moment.

AXES OF ROTATION

The aircraft has three Axes of Rotation. These are:

- **Lateral Axis**
- **Longitudinal Axis**
- **Vertical Axis**

Lateral Axis

Also known as the Pitch axis. A straight line passing through the CG parallel to the wing span. Pitch is controlled by the elevators. Motion about the Lateral Axis is referred to as Longitudinal Control or Longitudinal Stability.

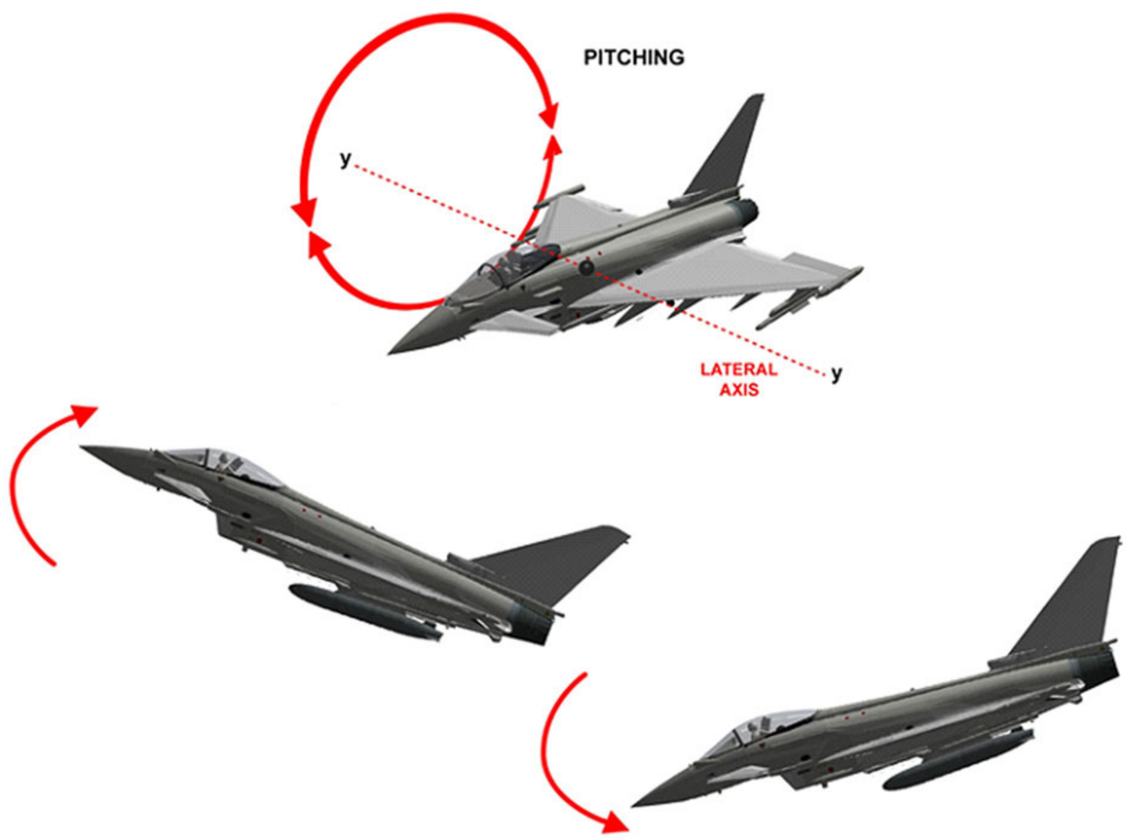

Fig 03. 2 - Pitching along the Lateral Axis

Longitudinal Axis

Also known as the Roll axis. A straight line through the fuselage of an aircraft passing through the C.of G. Roll is controlled by the ailerons. Motion about the Longitudinal Axis is referred to as Lateral Control or Lateral Stability.

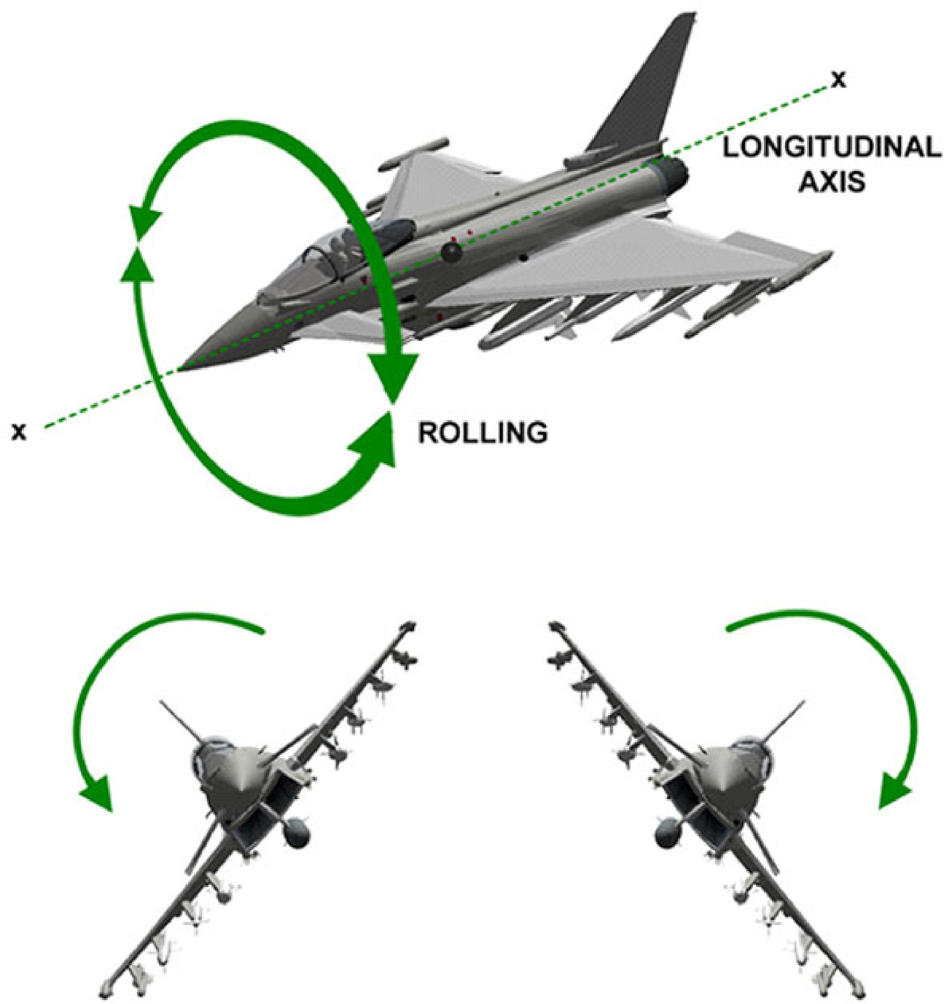

Fig 03. 3 - Rolling along the Lonitudinal Axis

Vertical Axis

Also known as the Yaw axis or normal axis. A straight line running vertically through the CG (the intersection of the Longitudinal and Lateral Axis). Yaw is controlled by the rudder. Motion about the Vertical Axis is referred to as Directional Control or Directional Stability.

The aircraft can rotate around all three axes simultaneously or it can rotate around just one axis. Stability of the aircraft is the combination of forces that act around these three axes to keep the pitch attitude of the aircraft in a normal level flight attitude, the wings level, and the nose of the aircraft directionally straight along the desired path of flight.

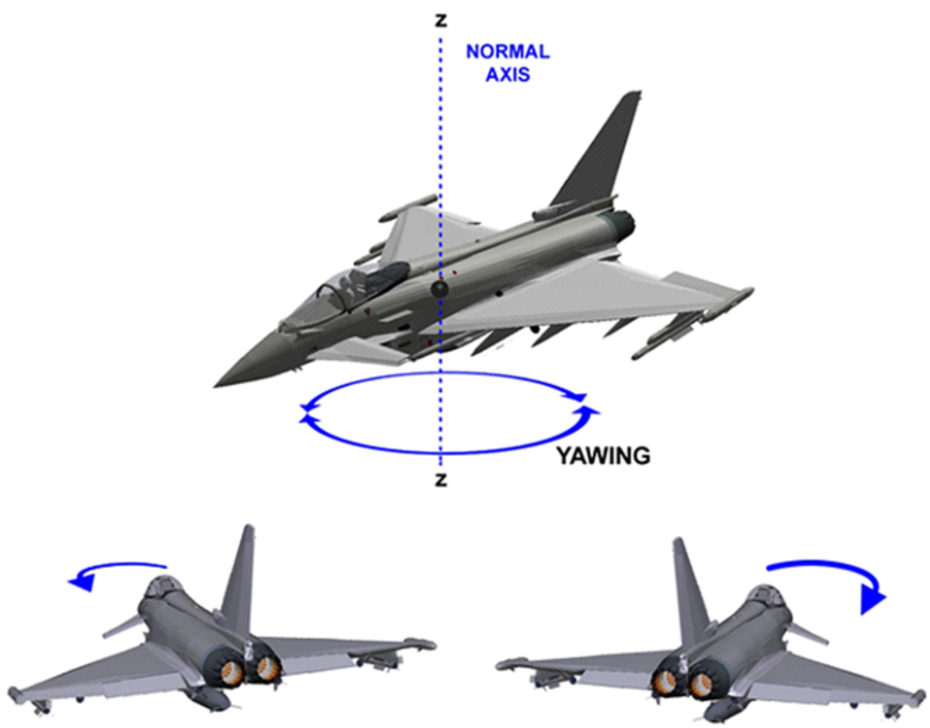

Fig 03. 4 - Yawing along the Vertical Axis

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Forces In Glide

If an aircraft's engine should fail, it will descend in a normal flying attitude; it is then said to be gliding. In a steady glide, the aircraft is kept in equilibrium by only three forces, Lift, Drag and Weight.

Glide Ratio

The ratio of every unit of vertical distance an aircraft descends, to the horizontal distance it travels while gliding. Large transport aircraft have a glide ratio of around 1: 18.

The Glide Angle

The total reaction (the resultant of the lift and drag) must be exactly equal and opposite to weight. The gliding angle is equal to the angle formed between the lift and the total reaction. The higher the lift - drag ratio, the smaller the gliding angle, thus increasing the glide ratio.

If an aircraft glides at an angle of attack either greater or less than the optimum lift - drag ratio, then in each case the path of decent will be steeper.

- **If greater than the optimum angle the speed will be below the correct gliding speed.**
- **If less than the optimum angle, the speed will be greater than the correct gliding speed.**

The glide angle depends on the ratio of lift to drag and is independent of weight. If the weight is greater there will need to be a corresponding increase in the total reaction and greater lift and drag but the proportions remain the same.

However greater lift and drag can only be obtained through greater speed. The distance covered will be the same but the time taken to descend will be reduced. In a glide, a component of weight balances the drag.

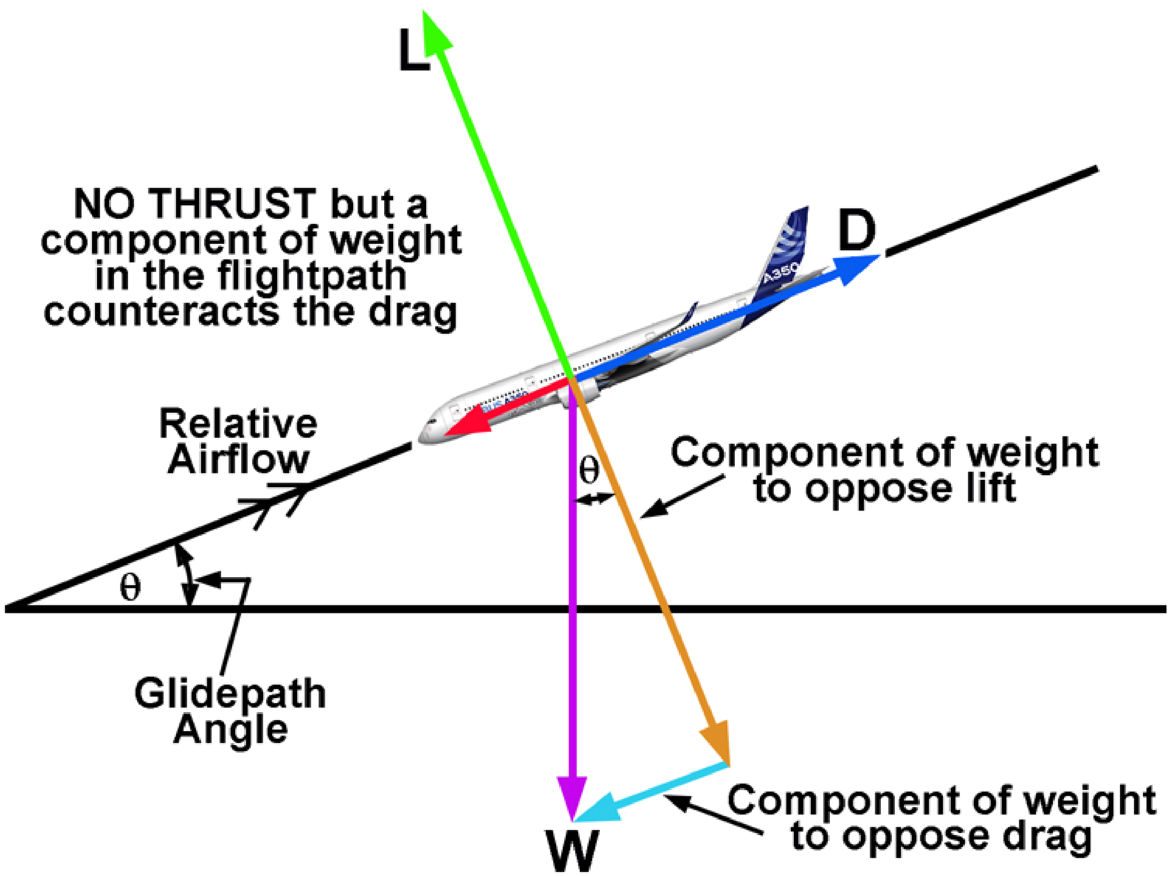

Fig 03. 5 - Forces in a Glide

FLIGHT PHASE

Take-Off

When a flight begins at T/0, the aircraft is loaded with enough fuel for its flight, plus the crew and passengers or cargo. At this point it may be at its heaviest weight, which is known as Maximum Take-off Weight (MTOW). The engines will be operating at their highest power settings and the aircraft will be accelerated along the runway.

Because we want the aircraft to use as short a run as possible, drag will be kept to a minimum. This means keeping the AOA as low as possible. When sufficient speed has been reached, the pilot will increase the angle of attack by raising the nose wheel off the ground. The force for this comes from the Horizontal stabiliser, which is producing negative lift.

A partial flap setting may be used to produce more lift at a lower speed.

Climb

A change in lift occurs when back elevator pressure is first applied. Raising the aircraft's nose increases the angle of attack and momentarily increases the lift. Lift at this moment is now greater than weight and starts the aircraft climbing.

After the flight path is stabilised on the upward incline, the angle of attack and lift revert to about the level flight values. Generally speaking, the forces of thrust and drag, and lift and weight, become balanced

The airspeed stabilises at a value lower than in straight and level flight at the same power setting. This is because a component of the aircraft's weight acts in the same direction as, and parallel to, the total drag of the aircraft, thereby increasing the total effective drag.

The ability to climb comes from the engines delivering enough thrust to overcome the total effective drag. In fact, because the thrust line is inclined uphill, a portion of thrust can be assigned to supporting the weight of the aircraft and less lift is required from the wings in a climb than in level flight.

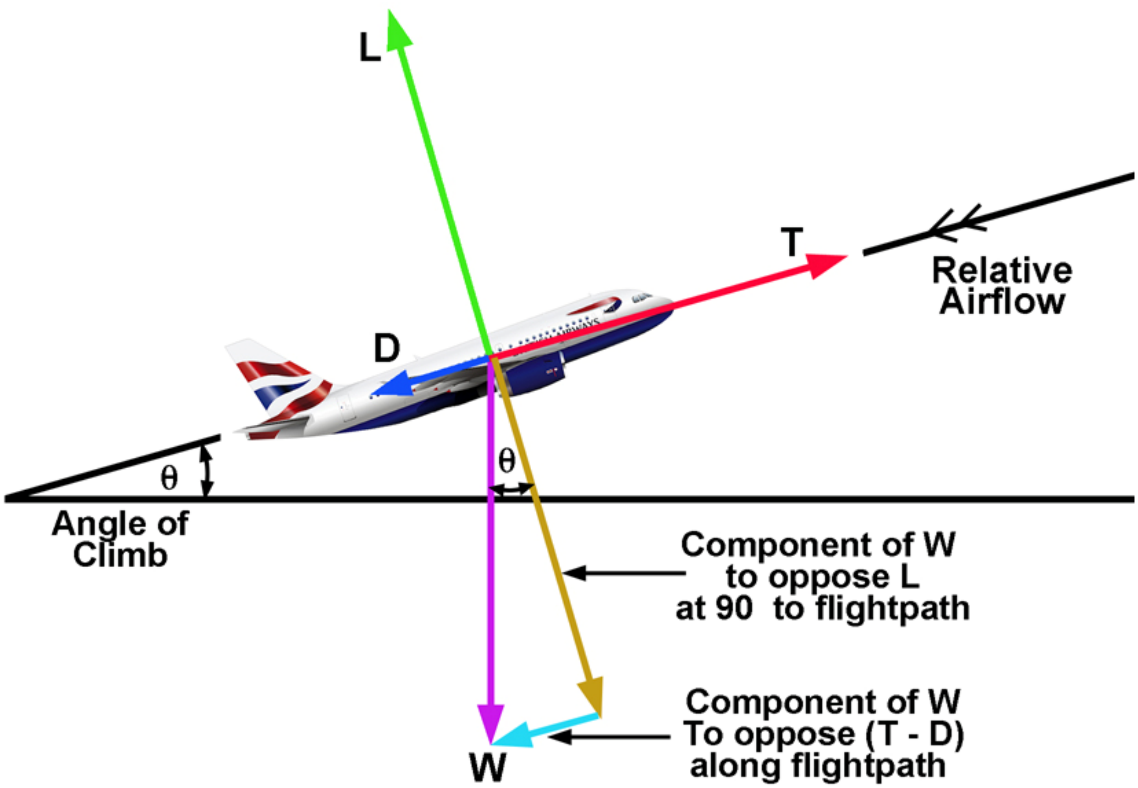

Fig 03. 6 - Aircraft in Climb.

Cruise

At the top of the climb the aircraft is levelled off. The power is reduced to the normal cruise setting and the four forces stabilise.

As fuel is burned the aircraft becomes lighter and less lift is required. The angle of attack can be reduced and unless thrust is reduced the aircraft will accelerate.

During cruise, the altitude normally remains constant, but the aircraft heading is changed frequently. This is done by rolling the aircraft in the desired direction or turn. When this happens the distribution of the four forces is changed.

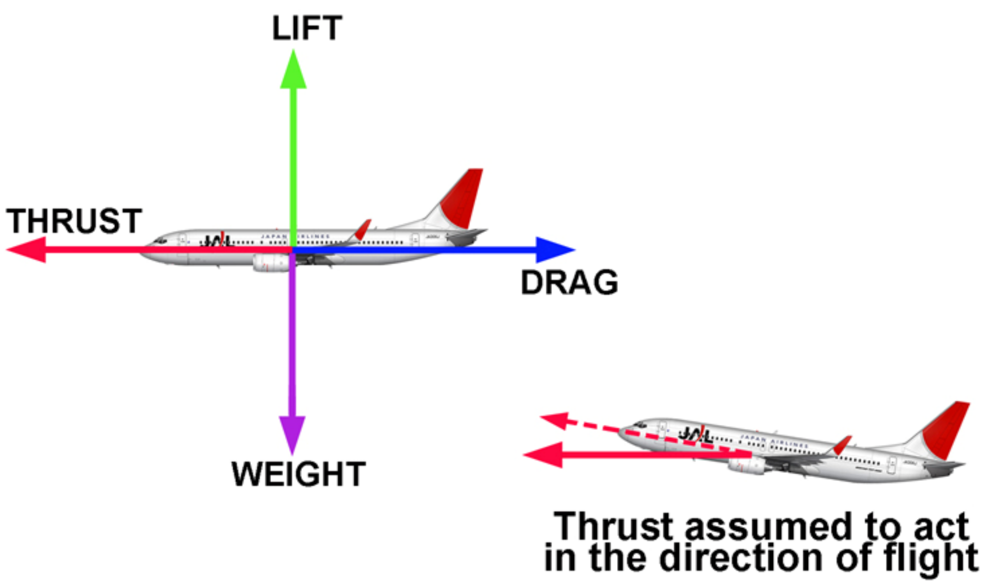

Fig 03. 7 - Level Flight (Cruise)

Descent

A descent, or glide, is a basic manoeuvre in which the aircraft is losing altitude in a controlled way with little or no engine power. Forward motion is maintained by gravity pulling the aircraft along an inclined path. The descent rate is controlled by the pilot balancing the forces of gravity and lift.

Turning

We look at the case in which the turn is made at constant altitude. Any time a turn is made by any vehicle, whether it is a bicycle, car, train or aircraft, a force must be provided towards the centre of the turn because there is an acceleration directed opposite the center.

In the case of a car or bicycle this force is provided by the tyres, in the case of a train it is provided by the rails. For an aircraft some other means must be found and this is done by tilting, or banking the aircraft so that a component of the lift force produced by the wings acts in the required direction

Thus the wings must produce a higher amount of lift than was required for normal straight and level flight. This extra lift means that, for a given speed, the wing must be operated at a higher angle of attack in the turn and, in addition, the increase in the lift will be accompanied by an increase in drag.

This drag will, in turn, mean that the power required to sustain a steady turn is greater than that required for flight in a straight line at the same speed. The angle of bank, lift, drag and required power all increase as the turn is tightened, and it may be that the minimum radius of turn which can be achieved is limited by the amount of power that is available from the engine.

Alternatively the demand for extra lift may cause the wing to stall before this point is reached, and stalling may therefore prove to be the limiting factor.

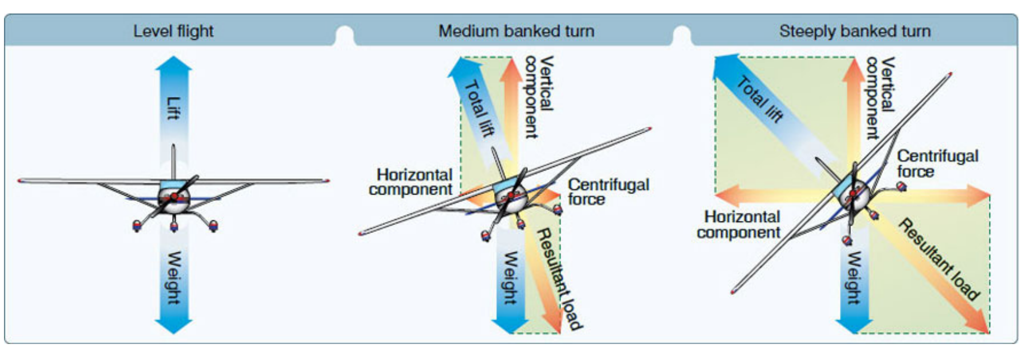

Fig 03. 8 - Turns

Wing Loading

This is the all up weight of the aircraft divided by the wing area, i.e.   the amount of the total weight carried by unit area of the wing It is usually given in kg/m^2.

Load Factor

The load factor is the ratio of the weight of the aircraft to the load imposed during a manoeuvre.

The load factor will change any time that the aircraft carries out a manoeuvre. This is because in straight and level flight at constant speed the aircraft is subject to 1g, any change to straight and level flight will change the G forces and therefore the load factor.

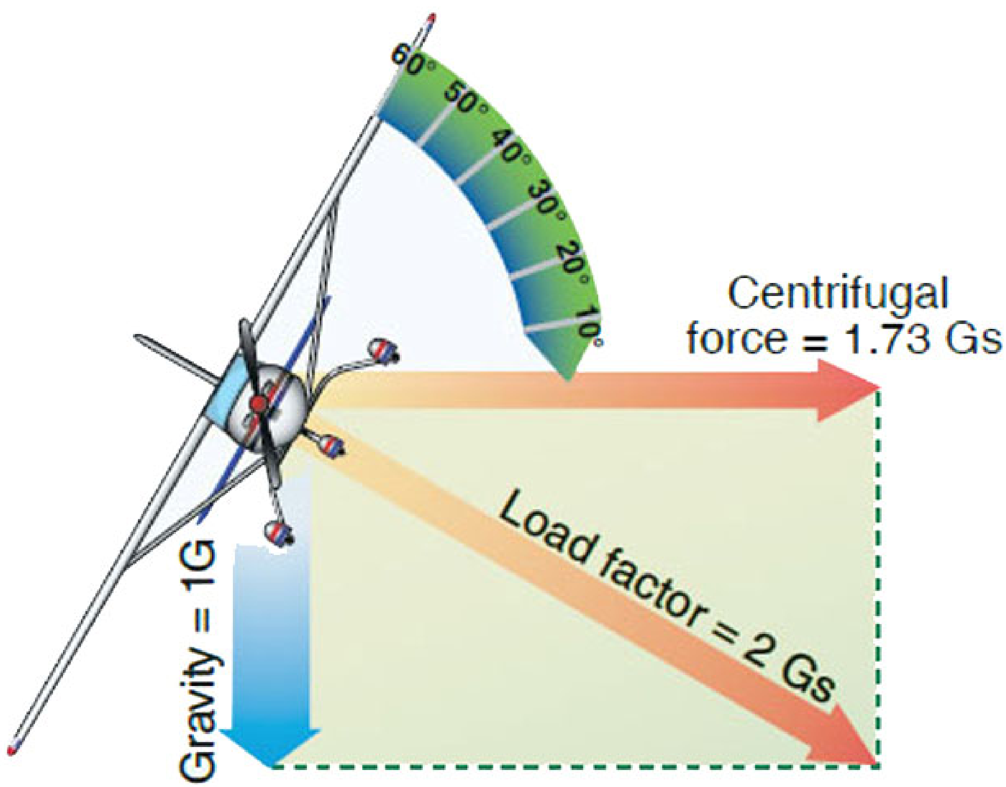

Fig 03. 9 -   Load factor

Effect of Weight (Wing Loading) on the Stall

Consider a wing which is producing sufficient lift to support a stated weight and is flying just below the stalling angle. If additional weight is placed on the wing the new weight is greater than the lift and level flight is no longer possible under the same conditions. To maintain level flight more lift must be produced and this can be done by increasing either the angle of attack or the speed.

Because the angle of attack in the example is almost at the stall, this cannot be increased without losing lift. It is therefore necessary to increase the speed to some higher figure until the lift again equals the weight. Therefore, although the angle of attack is still unchanged at a figure on the verge of the stall, the corresponding speeds, i.e. The stalling speed, is higher than the speed necessary at a lower wing loading.

To sum up, there is a minimum speed (the stalling speed) at which a wing can sustain a stated wing loading in level flight. The greater the wing loading the greater the stalling speed, and vice versa.

If the stalling speed, in level flight, is known for one weight, then the approximate stalling speed, at any other weight, can be calculated from the formula:

$$V_1 = V_2\sqrt{\frac{W_1}{W_2}}$$

where: V_1 = New Stalling Speed
V_2 = Original Stalling Speed
W_1 = New Weight
W_2 = Original Weight

It is interesting to note that the net effects of the additional Weight are exactly the same as the effects of an increase of altitude, i.e.:

- **Slight reduction in maximum speed.**
- **Large reduction in rate of climb.**
- **Increase in stalling speed.**

Lift Augmentation (High-Lift Devices)

High lift devices are used to provide extra lift when the aircraft is at low airspeeds and or high angles of attack. i.e. takeoff and landing. This is accomplished by changing the coefficient of lift and/or the wing area.

The main methods to achieve these changes are:

- **Wing Leading Edge Slots**
- **Wing Leading Edge Slats**
- **Wing Trailing Edge Flaps**

The purpose of most of them is the same: to increase the camber and or the surface characteristics of the wing so it can operate with a higher angle of attack without the airflow over the top surface becoming turbulent and breaking away.

When the flaps/slats are lowered and the camber is increased, both the lift and the drag are increased. Most flap/slat installations are designed in such a way that the first half of flap/slat extension increases the lift more than the drag, and partial flaps/slats are used for takeoff.

Lowering the flaps/slats all the way increases the drag more than the lift, and full flaps are used for landing.

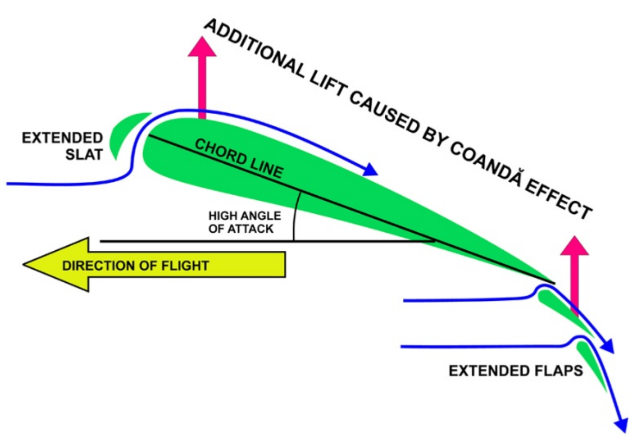

Fig 03. 10 - Airflow through Flaps/Slats

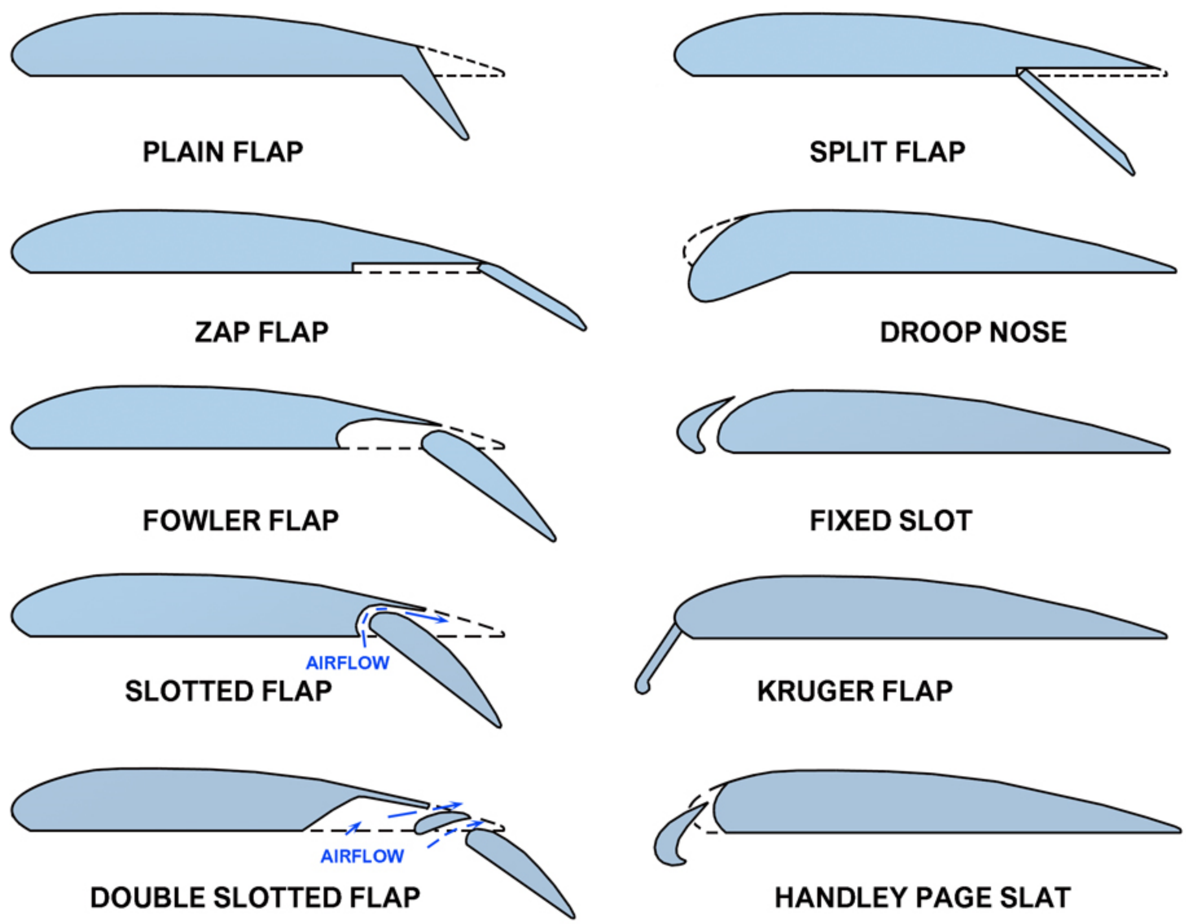

Fig 03. 11 - Types of Flaps/slots/slats

Effect of Flap on the Stalling Angle

When the trailing edge flap is lowered the angle of attack for level flight under the prevailing conditions is reduced. For each increasing flap angle there is a fixed and lower stalling angle of attack. The lower stalling angle is caused through the change in the aerofoil section when the flap is lowered.

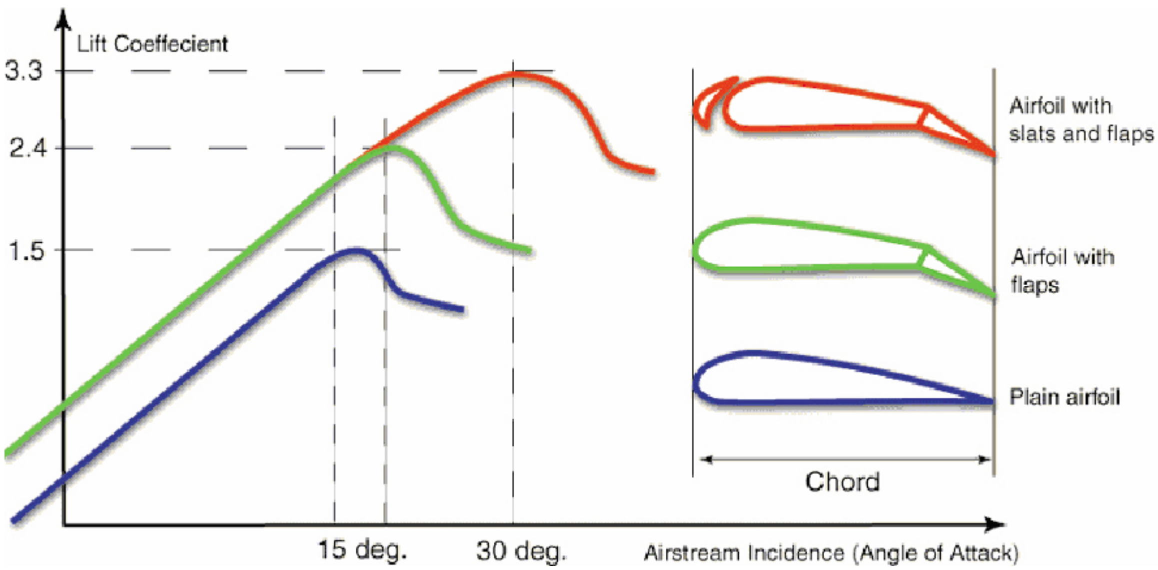

Fig 03. 12 - Lift Coefficient

Effect of Flap on the Lift / Drag Ratio

Lowering the flaps produces an increase in the lift coefficient at a given speed but at the same time the greater camber also causes an increase in the total drag. It can be stated that when the flaps are lowered to a given angle the lift / drag ratio is always reduced. The best lift / drag ratio with the flaps lowered is obtained with the flap at some angle between 15° and 35°, the exact angle depending principally on the aerofoil section used and the type of flap employed and its area.

For a typical trailing edge flap, as soon as the flap starts to lower, the lift and drag start increasing. For about the first third of range there is a steady rise in the C_L and this continues to increase at a reduced rate, and during the final part of the movement a further very small increase occurs.

In conjunction with the lift, the drag also increases, but the rate of increase during the first third of range is small compared with that which takes place during the remainder of the movement, the final third of range producing very rapid increase in the rate at which the drag has been rising. Since the flap is basically a means of increasing lift, the maximum angle rarely exceeds about 60°.

For landings, the high drag of the fully lowered flap is useful since it permits a steeper approach without the speed becoming excessive (i.e. it has the effect of an airbrake). The increased lift enables a lower approach speed to be used and the decreased stalling speed means that the touch down is made at lower speed. The high drag has another advantage in that it causes a rapid deceleration during the period of float after rounding out and before touching down.

LIST OF ILLUSTRATIONS

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SECTION 08-04 A1/B1/B2

HIGH SPEED AIRFLOW

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HIGH-SPEED AIRFLOW

In high-speed flight (faster than the speed of sound), the assumptions of air incompressibility used in low-speed aerodynamics no longer apply. In subsonic aerodynamics, lift is generated by the forces on a body moving through air. For example, at airspeeds below about 260 knots (480 km/h; 130 m/s; 300 mph), air can be considered incompressible in relation to an aircraft; at a fixed altitude, its density remains nearly constant while its pressure varies. Subsonic aerodynamic theory also assumes negligible effects of viscosity (the resistance of a fluid to motion) and classifies air as an ideal fluid, adhering to principles like continuity, Bernoulli's principle, and circulation. In reality, air is compressible and viscous. While these properties' effects are negligible at low speeds, compressibility effects become significant as airspeed increases.

TRANSONIC AND SUPERSONIC FLIGHT

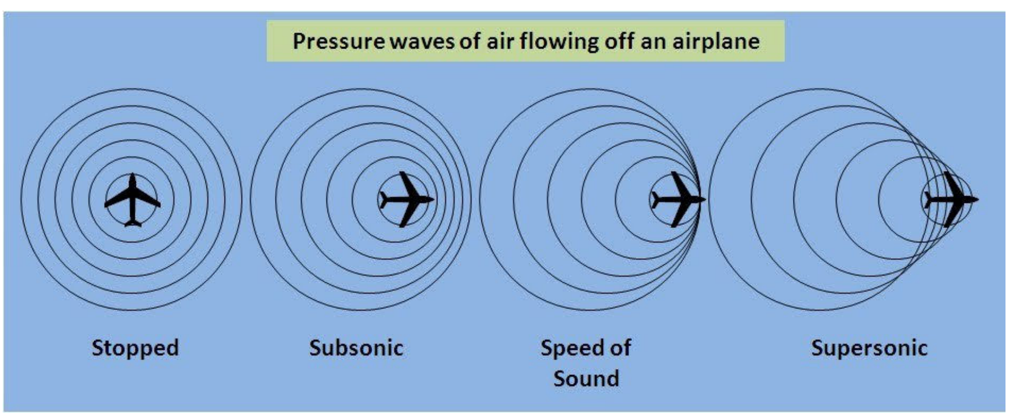
At speeds approaching the speed of sound, compressibility (and to a lesser extent, viscosity) becomes crucial. In transonic speed ranges, compressibility changes the air density around an airplane.

Subsonic Airflow: When air flows at subsonic speed through a diverging passage, its velocity decreases, and static pressure rises without changing density. In a converging passage, subsonic air speeds up, and static pressure decreases.

Supersonic Airflow: At supersonic speeds, air acts as a compressible fluid. In a converging passage, supersonic air slows down, and its pressure and density increase. Airplanes flying supersonic must have differently shaped wings due to these changes.

As an aircraft flies supersonic, it generates shock waves, creating a sonic boom along its path. From the aircraft's perspective, the boom seems to sweep backward.

During flight, a wing produces lift by accelerating the airflow over its upper surface, which can reach supersonic speeds even if the airplane is flying subsonic (Mach number < 1.0). At extreme angles of attack, the speed of the air over the wing may double the airplane's airspeed, resulting in both supersonic and subsonic airflows on the aircraft. When airflow reaches sonic speeds at certain locations, further acceleration causes compressibility effects such as shock wave formation, increased drag, buffeting, stability, and control issues. Subsonic flow principles are invalid above this point.



1. PRESSURE WAVES

THE SPEED OF SOUND

Sound, in relation to airplanes, is a series of pressure disturbances in the air, similar to waves created by dropping a rock in water. As an airplane flies, it creates sound energy in the form of pressure waves, which travel at the speed of sound (761 mph at a standard day temperature of 59°F). The speed of sound in air changes with temperature, increasing as temperature rises.

MACH NUMBER AND CRITICAL MACH NUMBER

Mach Number: The ratio between an object's speed and the speed of sound.

Critical Mach Number (Mcr): The lowest Mach number at which airflow over any part of the aircraft reaches the speed of sound. Approaching Mcr causes oscillatory forces (buffeting) felt as heavy vibrations. Exceeding the speed of sound smooths out the flight but causes shock waves.

FLIGHT REGIMES: SUBSONIC, TRANSONIC, AND SUPERSONIC

Subsonic Flight: All airflow around the airplane is below the speed of sound (Mach < 1). The critical Mach number varies with wing design but typically is just over Mach 0.8.

Transonic Flight: Part of the airplane experiences subsonic airflow, and part experiences supersonic airflow, forming shock waves that cause stability problems and airflow separation from the wing. Transonic speeds range from Mach 0.80 to 1.20.

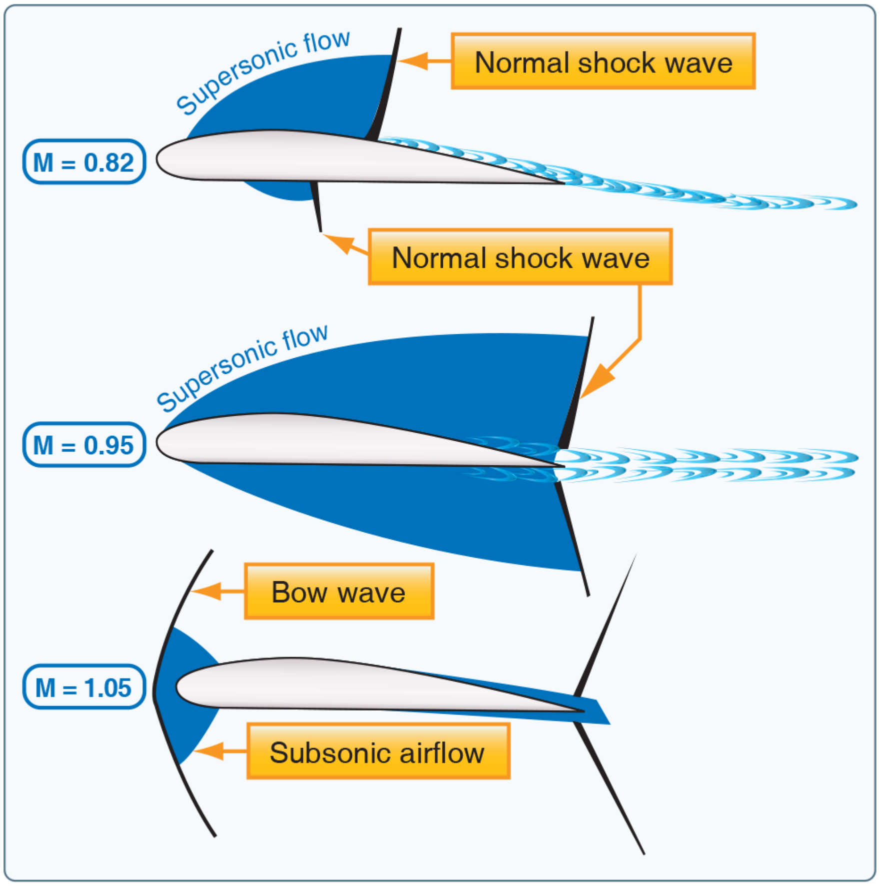
Supersonic Flight: The entire airplane experiences supersonic airflow. Shock waves attach to the wing's trailing edge, and the speed ranges from Mach 1.20 to 5.0. Hypersonic flight (Mach > 5) includes all supersonic speeds but is more extreme.

SHOCK WAVES

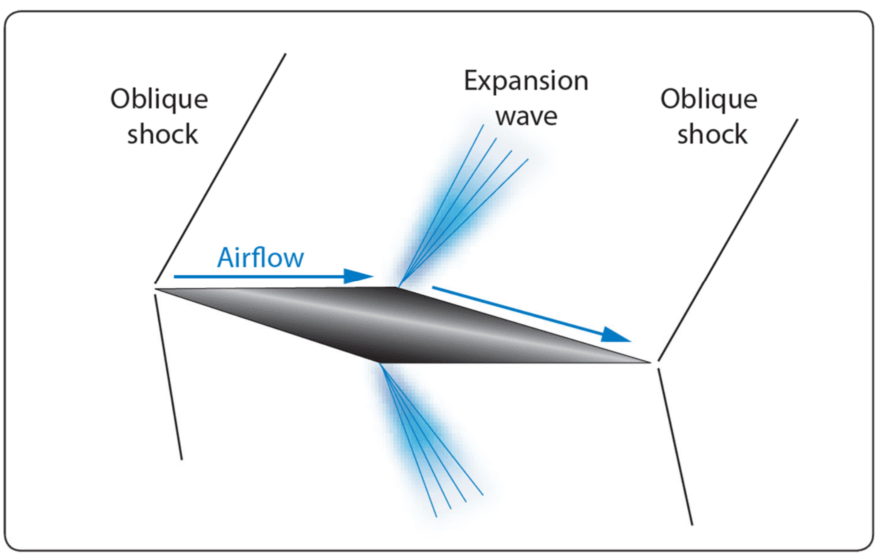
Normal Shock Waves: Form perpendicular to the airflow during transonic and supersonic flight, resulting in subsonic airflow behind the wave with higher static pressure and density.

Oblique Shock Waves: Form on sharp-edged surfaces of supersonic aircraft, maintaining supersonic airflow behind the wave but with increased static pressure and density.

Expansion Waves: Occur when supersonic air expands and changes direction, increasing velocity and decreasing static pressure and density.



2. NORMAL SHOCK WAVES



3. OBLIQUE & EXPANSION SHOCK WAVES

HIGH-SPEED AIRFOILS

Transonic Flight Challenges: Subsonic airfoils experience control and stability changes as airflow transitions between subsonic and supersonic. Six scenarios illustrate this transition:

Entire wing experiences subsonic airflow.

Airflow over the wing reaches Mach 1.

Normal shock wave forms on the wing's top.

Serious airflow separation occurs behind the shock wave.

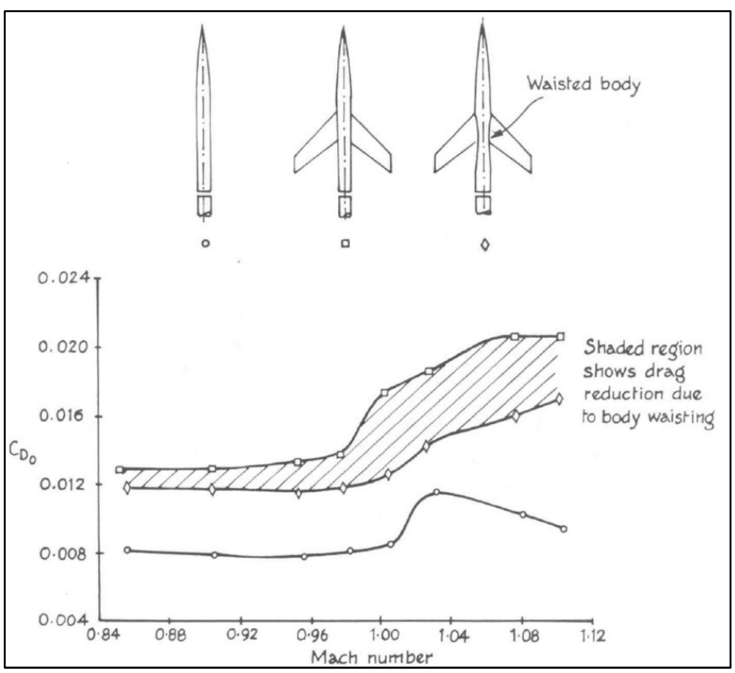
Shock waves move to the wing's trailing edge.

A new shock wave forms just forward of the wing's leading edge.

WHITCOMB’S AREA RULE

The area rule is the result of a series of experiments done by Dr. Richard Whitcomb, which was (even now) the most intuitive approach to one of the challenging problems in the field of aerodynamics.

In a transonic flight, when the airspeed is close to the transonic range, the local airflow around some parts of the airplane, commonly in the upper surface of the wings, tends to reach Mach 1 easily. Such localized supersonic flows produce shockwaves, which significantly affect the performance of the aircraft by producing a form of a rapid increase in drag – called as the wave drag. The area rule is a very important rule that addresses this effect by minimizing any rapid changes in the longitudinal cross-sectional area of an aircraft by making appropriate design adjustments.



4. WHITCOMB’s AREA RULE MODELS

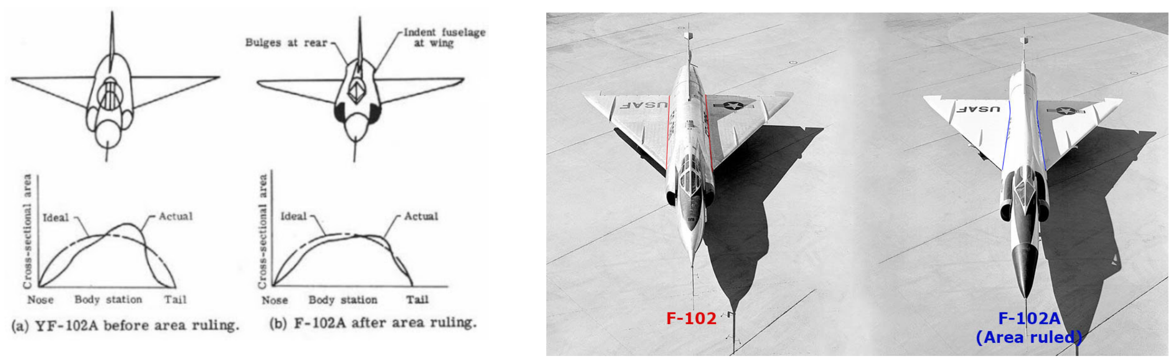
How does Whitcomb's area rule work?

Whitcomb observed the shockwave formations from various wind tunnel models involving shapes that included wings and just the fuselage and the indented shape as shown below.

Whitcomb found that the drag was proportional to the discontinuity in the cross-sectional area throughout the length of the airplane. He concluded that the presence of wing adds an extra volume at that point and by indenting and reducing the volume of the fuselage it could lead to a smoother area distribution, which in turn would reduce the wave drag. The two shapes in the middle reflect the identical cross-sectional area distribution.

The very first application of the area rule was in the modification of the Convair F-102 Delta Dagger as F-102A. The F-102 performed poorly due to high wave drag and failed to achieve a supersonic flight. The aircraft was then redesigned as F-102A by reducing (indenting) the area of the fuselage at the waist, following Whitcomb's area rule. The F-102A was later able to achieve Mach 1.22, with a considerable reduction in the transonic wave drag.

Here is a schematic cross-sectional area distribution plot for both the variants of the aircraft.



5. F-102A AREA RULE APPLICATION

Note that the second plot on the right has a smoother distribution with respect to the contour where the fuselage is indented.

So by indenting the fuselage (reducing the volume of the fuselage) at the waist, the cross-sectional area is maintained as smooth as possible, providing almost a uniform and continuous area of smooth interaction to the airflow. Thus by doing this, one can achieve an area distribution close to a Sears-Haack body, which is the theoretical design for a fuselage with the lowest possible wave drag for a given volume.

However, the Whitcomb's area rule is only good for reducing the wave drag due to the wings, and it doesn't really get rid of all the wave drag anyway. Other forms of area ruling with configurations like the tail-mounted engines, specifically placed streamlined components are still implemented today.

A very familiar example of this area ruling can be seen in the design of the Boeing 747 aircraft. The hump in the front is the result of applying the area rule to maintain a streamlined cross-sectional area for the near transonic flights. In the case of modern supersonic designs, as the engines turned more powerful, the area rule is less considered, mostly favoring the cost-efficiency in manufacturing.

AERODYNAMIC HEATING

High-speed flight generates significant heat due to air friction. The SR-71 Blackbird, flying at Mach 3.5, experiences skin temperatures from 450°F to over 1,000°F, necessitating titanium alloy construction. The Concorde reduced its cruise speed from Mach 2.2 to Mach 2.0 due to structural issues from aerodynamic heating. Future hypersonic aircraft must address structural stresses caused by heat.

THE IMPACT OF SHOCK WAVES ON AIRCRAFT ENGINE INTAKES

Shock waves have a profound impact on aircraft engine intakes, particularly in high-speed and supersonic flight regimes. Here are the key effects and considerations related to shock waves in this context:

1. Pressure Increase

Normal Shock Waves: These shock waves occur perpendicular to the flow direction and result in a sudden and significant increase in pressure, temperature, and density while decreasing the Mach number to subsonic speeds. The abrupt pressure rise can cause potential issues for engine performance and structural integrity if not properly managed.

Oblique Shock Waves: These occur at an angle to the airflow, resulting in a more gradual pressure increase compared to normal shock waves. Managing the angle and positioning of these waves is crucial for efficient compression and minimizing drag.

2. Temperature Rise

The passage of airflow through shock waves results in a considerable temperature increase. High temperatures can impact material properties, lead to thermal stresses, and challenge cooling systems. Managing this temperature rise is essential to prevent damage to engine components.

3. Flow Deceleration

Shock waves cause a sudden deceleration of the airflow. This deceleration is necessary to transition from supersonic to subsonic speeds before the air enters the engine compressor, which typically operates more efficiently at lower speeds. However, excessive deceleration can lead to inefficiencies and potential flow separation.

4. Energy Losses

The energy losses associated with shock waves result in a decrease in total pressure across the shock. This loss of total pressure reduces the overall efficiency of the engine and must be minimized through careful design and management of the intake system.

i. Pressure Increase Across a Shock Wave:

Static Pressure Increase: When airflow encounters a shock wave, particularly a normal shock wave, the static pressure increases significantly. This is because the airflow is suddenly compressed, leading to a rise in both pressure and temperature. For jet engines, especially at high speeds, increasing the static pressure is necessary to compress the incoming air before it enters the engine's compressor. This initial compression helps the compressor operate more efficiently.

ii. Total Pressure Decrease Across a Shock Wave:

Total Pressure Loss: Despite the increase in static pressure, there is a loss in total pressure across the shock wave. This is because shock waves are non-isentropic processes, meaning they involve energy dissipation. The total pressure after the shock is lower than before because some of the energy is converted into heat and other forms of irreversible losses (entropy increase). The decrease in total pressure represents a loss of useful energy that could have been converted into thrust. This loss reduces the overall efficiency of the engine because the engine must work harder to achieve the necessary compression and airflow for combustion.

iii. Design Optimization:

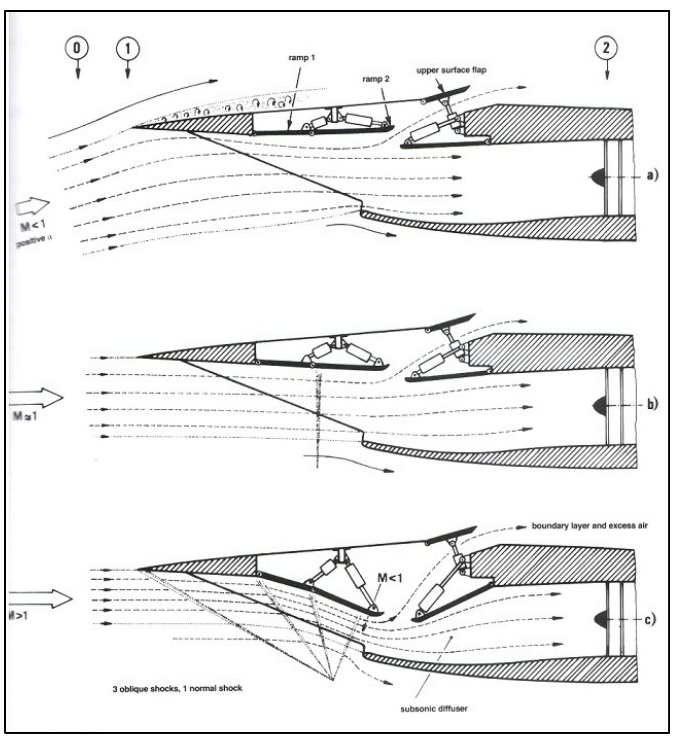
Engineers aim to design intakes that manage shock waves in such a way that maximizes static pressure recovery while minimizing total pressure loss. This involves placing shock waves strategically (e.g., using oblique shocks where possible) and using variable geometry to adapt to different flight conditions.

5. Shock-Induced Boundary Layer Separation

The interaction of shock waves with the boundary layer (the thin layer of air close to the intake surface) can lead to boundary layer separation. This separation can cause significant distortions in the airflow, reducing the efficiency of the intake and potentially leading to compressor stall or surge.

6. Intake Design Strategies

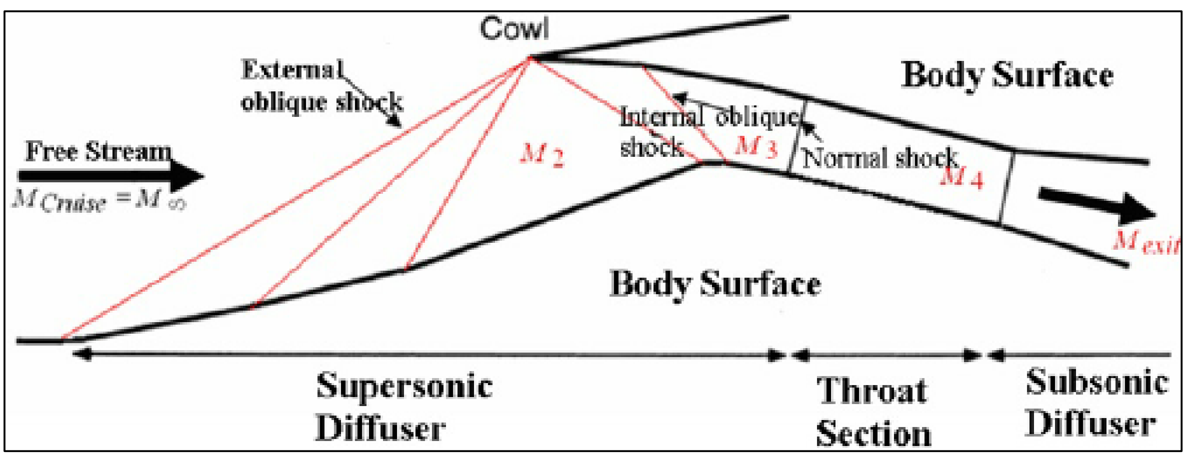
Ramp Intakes: For supersonic aircraft, ramps or compression surfaces are often used to generate controlled oblique shock waves. These ramps can be adjustable (variable geometry) to optimize shock wave positioning and strength for different flight speeds and conditions.



6. RAMP INTAKE

Conical Intakes: Some designs use a cone (or centerbody) to generate oblique shock waves, progressively slowing and compressing the airflow.

Mixed-Compression Intakes: These combine features of both external and internal compression, using a combination of oblique and normal shocks to achieve the desired flow characteristics.



7. CONICAL INTAKE

7. Shock Position and Stability

Maintaining stable shock wave positions is crucial. Unstable shock waves can oscillate, leading to unsteady pressures and potential engine surges. Intake designs must ensure that shock waves remain at the desired location under varying flight conditions.

8. Supersonic Inlet Buzz

This is a phenomenon where shock waves oscillate within the intake, causing pressure fluctuations and potentially leading to engine instability. It is more likely to occur at off-design conditions and must be mitigated through proper intake design and control mechanisms.

9. Noise and Vibration

Shock waves can generate significant noise and vibration, impacting both the structural integrity of the intake and the overall acoustic environment of the aircraft. Effective design and damping mechanisms are necessary to mitigate these effects.

10. Thermal and Structural Stress

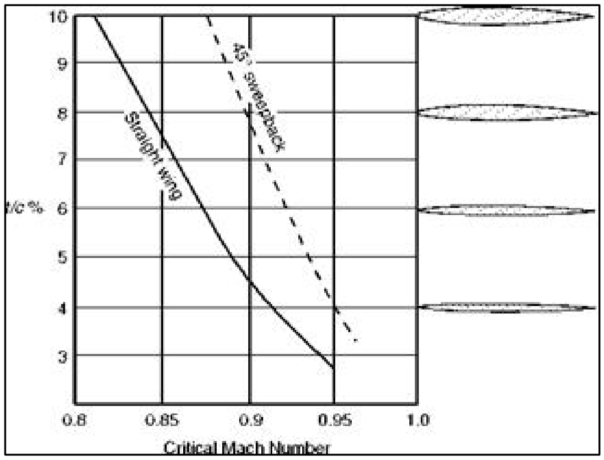
The temperature and pressure changes associated with shock waves impose thermal and structural stresses on the intake materials and structure. The design must account for these stresses to ensure long-term durability and performance.

Conclusion

Shock waves play a critical role in the performance of engine intakes in high-speed aircraft. They affect pressure, temperature, flow stability, and overall efficiency. Effective management of shock waves through advanced intake designs, such as ramps, cones, and mixed-compression systems, is essential for optimizing performance and ensuring the reliability and safety of supersonic and hypersonic aircraft.

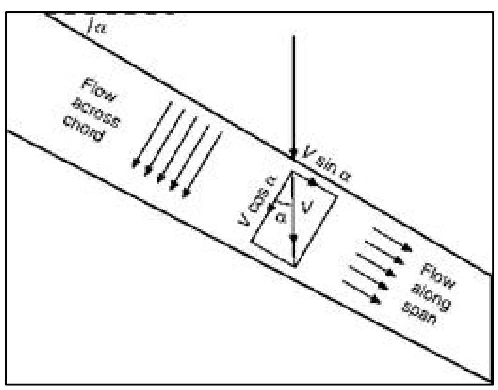
SWEEPBACK - RAISING THE CRITICAL MACH NUMBER

A main way of raising the critical Mach Number (and this applies only to the wings, tail, fin and control surfaces) is sweepback – not just the few degrees of sweepback that was sometimes used, rather apologetically and for various and sometimes rather doubtful reasons, on subsonic aircraft, but 40°, 50°, 70° or more.



8. SWEEPBACK-CRITICAL MACH NO

Sweepback of this magnitude not only delays the shock stall but reduces its severity when it does occur. The theory behind this is that, it is only the component of the velocity across the chord of the wing (V cos a) which is responsible for the pressure distribution and so for causing the shock wave (Fig. 9); the component У sin a along the span of the wing, causes only frictional drag. This theory is borne out by the fact that when it does appear the shock wave lies parallel to the span of the wing, and only that part of the velocity perpendicular to the shock wave, i. e. across the chord, is reduced by the shock wave to subsonic speeds.

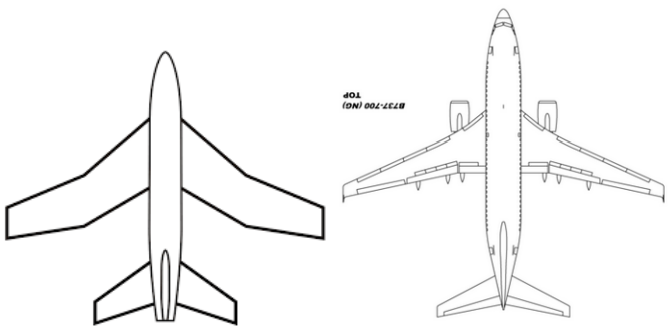


9. SWEEPBACK – COMPONENTS OF VELOCITY

As the figure clearly shows, the greater the sweepback the smaller will be the component of the velocity which is affected, and so the higher will be the critical Mach Number, and the less will be the drag at all transonic speeds of a wing of the same tic ratio and at the same angle of attack.

Experiment confirms the theoretical advantages of sweepback, though the improvement is not quite so great as the theory suggests. The dotted line in Fig. 8 shows how a wing swept back at 45° has a higher critical Mach Number than a straight wing at all values of t/c ratio, the advantage being greater for the wings with the higher values of t/c.

Of course, as always, there are snags, and the heavily swept-back wing is no exception. There is tip stalling – an old problem, but a very important one; in the crescent-shaped wing (Fig. 10) an attempt has been made – with some success – to alleviate this by gradually reducing the sweepback from root to tip.



10. CRESCENT-SHAPED WING

Additionally, CLmax is low, and therefore the stalling speed is high, and CLmax is obtained at too large an angle to be suitable for landing – another old problem, and one that can generally be overcome by special slots, flaps or suction devices.

There are also control problems of various kinds, and the designer doesn't like the extra bending and twisting stresses that are inherent in the heavily swept-back wing design. But whatever the problems sweepback seems to have come to stay – at least for aircraft which are designed to fly for any length of time at transonic or low supersonic speeds.

LIST OF ILLUSTRATIONS

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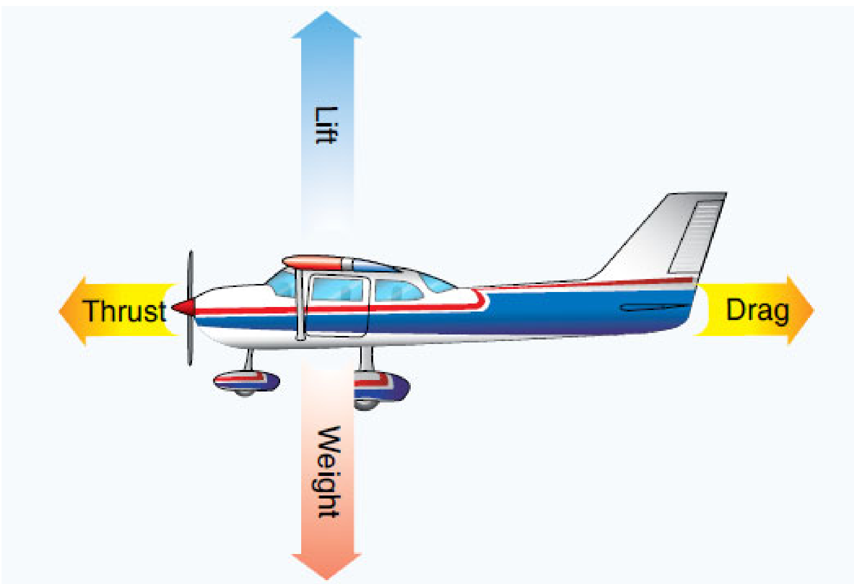

SECTION 08-05 A1/B1/B2

FLIGHT STABILITY AND DYNAMICS

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FLIGHT STABILITY

Flight Stability and Dynamics

The stability of an aircraft means its ability of return to some particular condition of flight (after having been slightly disturbed from that condition) without any efforts on the part of the pilot.

An aircraft may be stable under some conditions of flight and unstable under other conditions. Control means the power of the pilot to manoeuvre the aircraft into any desired position.

As already stated, an aircraft which, when disturbed, tends to return to its original position is said to be stable. If on the other hand, it tends to move farther away from the original position, it is unstable.

But it may tend to do neither of these and prefer to remain in its new position. This is called neutral stability, and is sometimes a very desirable feature.

Static Stability

This means that when disturbed from its flight path, forces will be activated which will initially tend to return the aircraft to its original position.

The returning forces may be so great that the aircraft will pass beyond the original position and continue in that direction until stability again tries to restore the aircraft into its original position. This will continue with the oscillations either side of the original becoming larger.

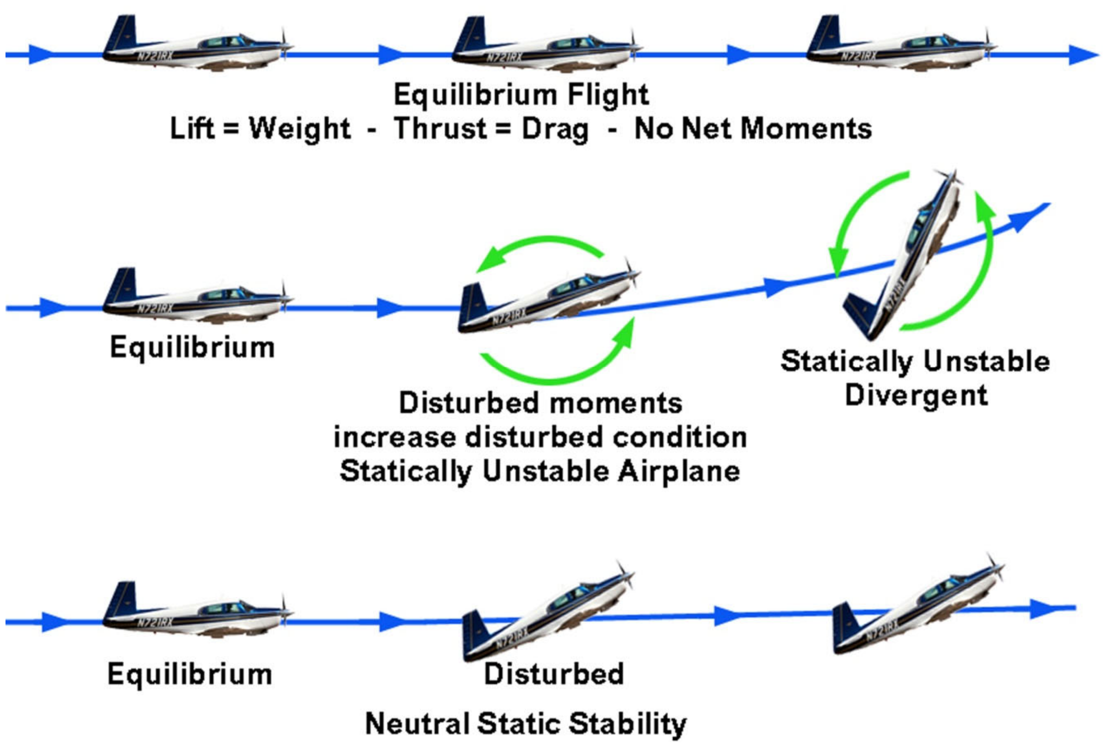

Fig 04. 1 - Static Stability

Dynamic Stability

Static stability, as was stated, creates a force that tends to return the aircraft to its original angle of attack, but it is the dynamic stability of the aircraft that determines the way it will return. It is concerned with the way the restorative forces act with regard to time.

In other words, it is the property which dampens the oscillations set up by a statically stable aircraft.

When the oscillations become smaller and eventually return the aircraft to its original position the aircraft has positive static and positive dynamic stability.

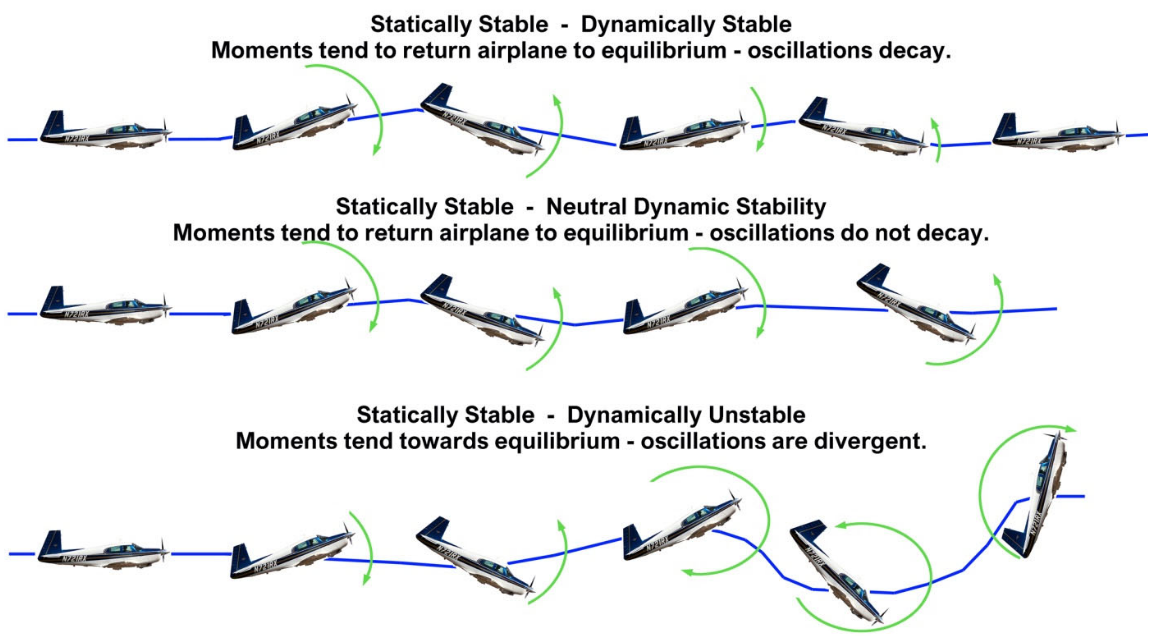

Fig 04. 2 - Static and Dynamic stability

Longitudinal Stability About The Lateral Axis

There are two important factors which will affect Longitudinal Stability. These are:

- **The Position of the Centre of Gravity**
- **The Pitching Moment on the Wings.**

The Position of the Centre of Gravity

This must never be too far back or a strong nose up tendency will exist. This is the most important consideration and is controlled by correct aircraft loading.

The Pitching Moment on the Wings

If the aircraft is disturbed nose up, the angle of attack is increased. This causes the center of pressure to move forwards. The effect of this is to pull the nose up further away from its original position. This is static instability which can be controlled by wing design but will never completely disappear.

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The Horizontal Stabiliser

In the same situation with the nose displaced upwards, the angle of attack of the horizontal stabiliser will alter. This will change the force produced by the stabiliser. This change of force will be a restoring moment.

The amount of restoring moment will depend on:

- **The Area of the stabiliser:**
  - **A larger stabiliser will supply more force.**
- **The Length of the Fuselage:**
  - **For a given size of stabiliser, a longer fuselage will give more leverage and more restoring moment.**

The nose-down force caused by the CG's position ahead of the center of lift is fixed and does not change with airspeed. But the tail load is speed dependent the higher the airspeed, the greater the downward force on the tail.

If the aeroplane is trimmed for level flight with the pilot's hands off the controls, and a wind gust causes the nose to drop, the aeroplane will nose down and the airspeed will increase.

As the airspeed increases, the tail load increases and pulls the nose back to its level flight condition. If the nose is forced up, the airspeed will drop off, and the tail load will decrease enough to allow the nose to drop back to level flight.

The flying wing aeroplane usually has a large amount of sweepback, and since they have no tail, their longitudinal stability is produced by washing out the tips of the wing. The speed-dependent downward aerodynamic force at the wing tip is behind the center of lift, and it produces the same stabilising force as that produced by a conventional tail.

Fig 04. 3 - Horizontal Stabiliser

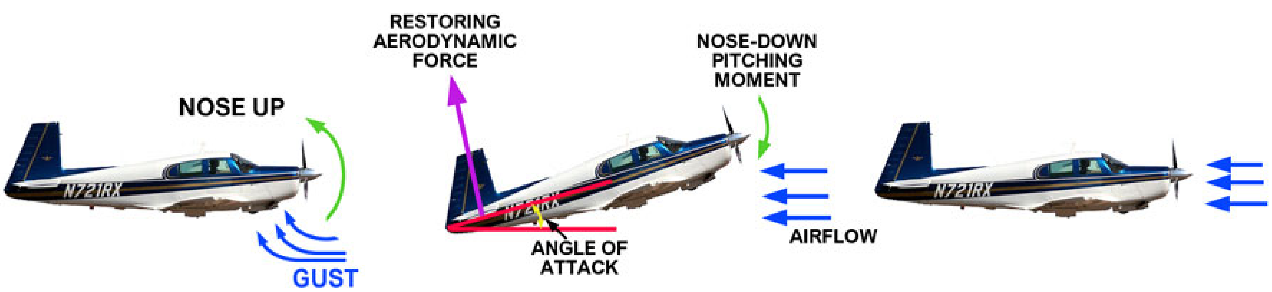

Fig 04. 4 - Longitudinal Stability

Lateral Stability About The Longitudinal Axis

An aircraft that tends to return to wings level flight after it has been disturbed is considered to be laterally stable.

The many factors that affect lateral stability are:

Dihedral Angle

This is the upward and outward slope of the wing. If the wing tip is higher than the wing root relative to the horizontal plane, the aircraft has positive dihedral. Negative dihedral is termed anhedral.

When an aircraft rolls, the lower wing presents a larger span as seen from the direction of the approaching air, the effect is to roll the aircraft back towards the horizontal. This will always be a restoring moment.

If the restoring moment is insufficient to restore level flight the aircraft will begin to sideslip.

Now the effect of dihedral comes into effect. In a sideslip the lower wing has in increased angle of attack while the upper wing has a reduced angle of attack.

The greater lift on the lowered wing will restore level flight.

Fig 04. 5 - Dihedral Angle

Sweepback

Sweepback Angle is the angle at which the wing points backwards from the root to the tip.

Sweepback is used mainly on high-speed aircraft and its primary purpose is to delay the formation of sonic shock waves which are produced at high speeds and cause a large increase in drag.

The secondary effect of sweepback is to improve lateral stability. When a side-slip occurs, the lower wing presents a larger span as seen from the direction of the approaching air, and as with dihedral, the effect is to roll the aircraft back towards the horizontal.

In general, as the sweepback angle is increased the dihedral angle will be reduced.

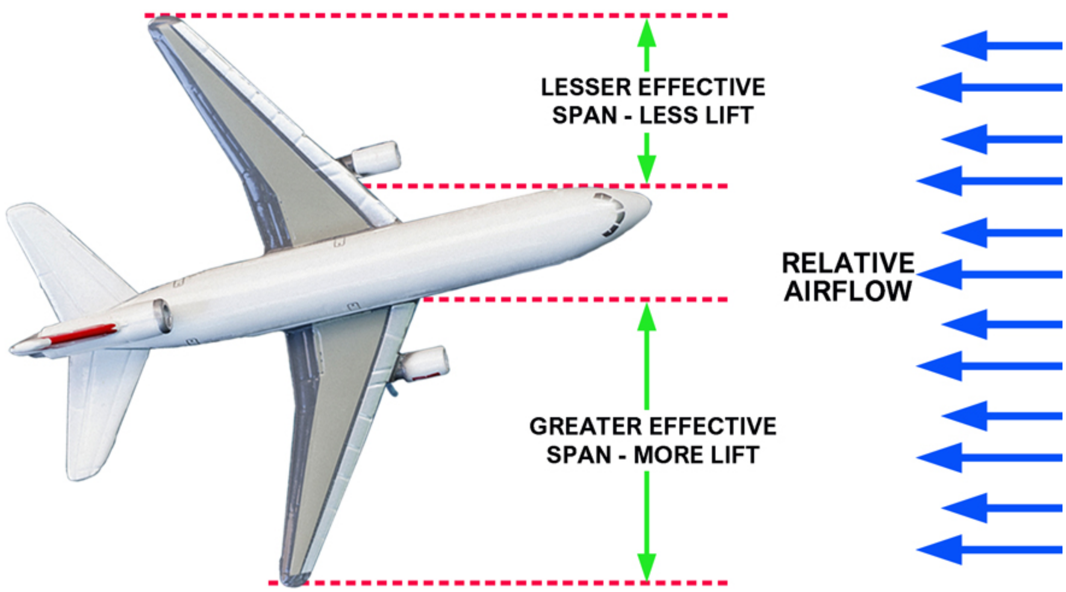

Fig 04. 6 - Sweepback and Lateral Stability

Keel Effect

If an aircraft begins to sideslip, all of the keel surface above the center of gravity will be presented to the relative airflow which will give a force to help in correcting the roll.

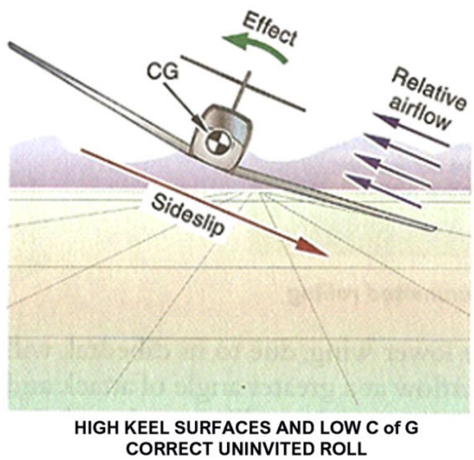

Fig 04. 7 - Keel Effect

High Wings

Mounting the wings above the center of gravity aids lateral stability because the weight of the aircraft will act as a pendulum to restore level flight.

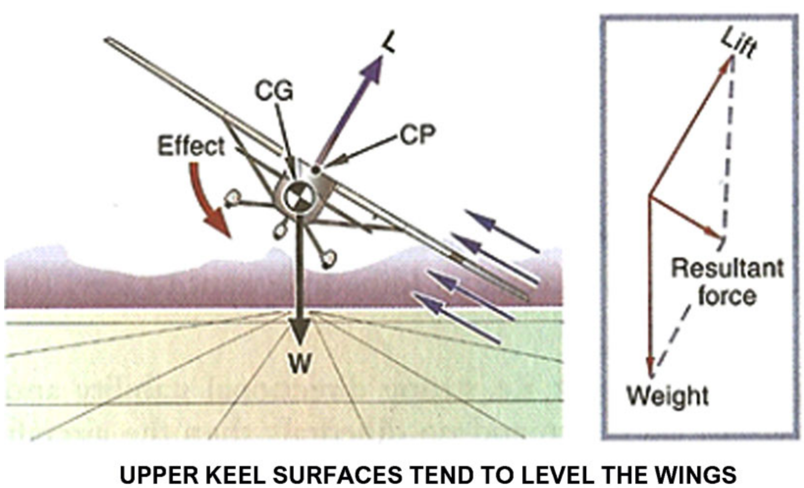

Fig 04. 8 - Upper Keel Effect

Directional Stability about the Vertical Axis

Directional Stability is displayed around the vertical axis and depends to a great extent on the quality of lateral stability. If the longitudinal axis of an aircraft tends to follow and parallel the flight pattern of the aircraft through the air, whether in straight flight or curved flight, that aircraft is considered to be directionally stable.

Directional stability is accomplished by placing a vertical stabiliser or fin to the rear of the centre of gravity on the upper portion of the tail section.

The surface of this fin acts similar to a weather vane and causes the aircraft to weathercock into the relative wind. If the aircraft is yawed out of its flight path, either by pilot action or turbulence, during straight flight or turn, the relative wind would exert a force on one side of the vertical stabiliser and return the aircraft to its original direction of flight.

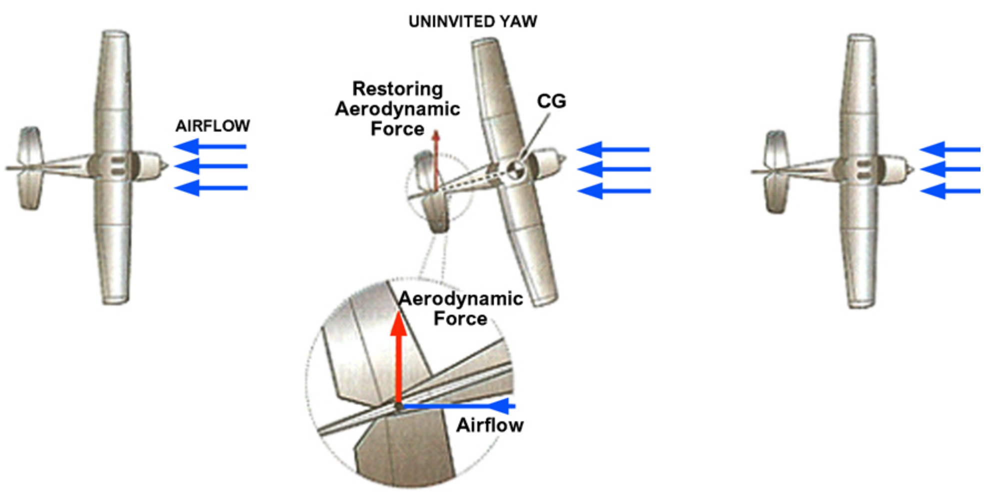

Fig 04. 9 - Directional Stability

Wing sweepback aids in directional stability. If the aircraft is rotated about the vertical axis, the aircraft will be forced sideways into the relative wind.

Because of sweepback this causes the leading wing to present more frontal area to the relative wind than the trailing wing. This increased frontal area creates more drag, which tends to force the aircraft to return to its original direction of flight.

LIST OF ILLUSTRATIONS